

HCDE 511

Information Visualization

Winter 2026

8

Week



Week

Quickly discuss A3

Trees & Networks

Break

Peer User Research

Project Progress Check-in (time permitting)

Project Group Work Time

Week 1

Intro
Data
encoding &
mapping

2

Graphical
excellence
& integrity
Critiquing

3

Fundamen-
tal graphs
Working
with data

4

Tableau &
dashboard
design

Visual
perception

5

Visual
perception,
cont.
Tasks &
inter-
actions

6

Collabora-
tive
critique

7

Multi-
variate
analysis

8

Trees &
Networks

Project
check-in

9

Animation,
context,
story-
telling,
different-
iation

10

Final pre-
sentations

Finals

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

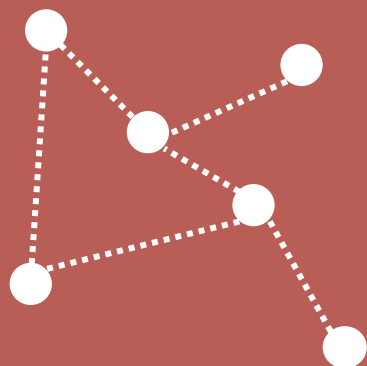
Project proposal

Rev. prop.

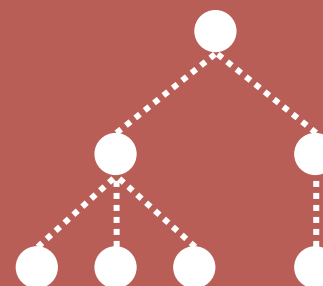
Final project deliverable

Final project writeup

Trees & Networks



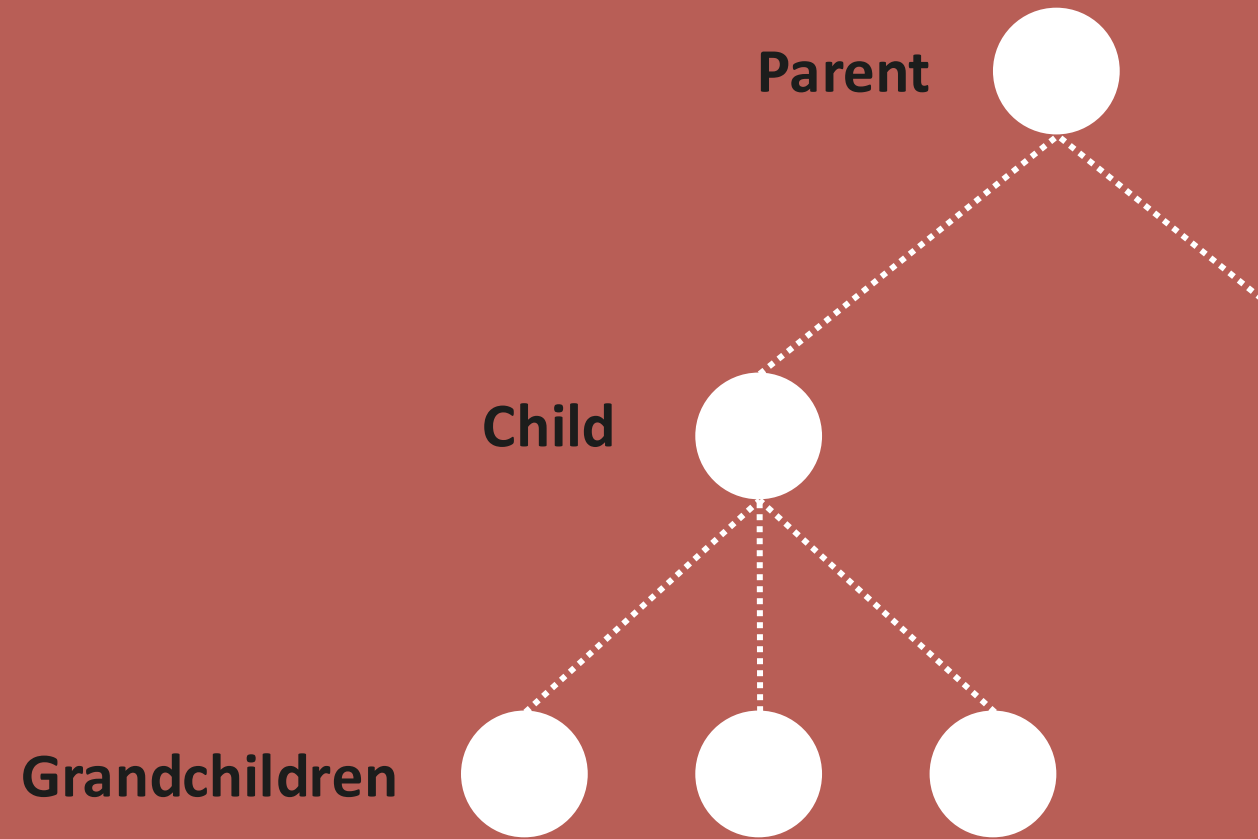
Networks



Trees

Why?

Representations of data with natural hierarchy



2011 NCAA® Division I Men's Basketball Championship

First Round
MARCH 15-16Second Round
MARCH 17-18Third Round
MARCH 19-20Regional
Semifinals
MARCH 24-25Regional
Finals
MARCH 26-27National
Semifinals
APRIL 2FILL OUT YOUR BRACKET
NATIONAL BRACKET DAY
MARCH 14National
Semifinals
APRIL 2Regional
Finals
MARCH 26-27Regional
Semifinals
MARCH 24-25Third Round
MARCH 19-20Second Round
MARCH 17-18First Round
MARCH 15-16

First Four

Dayton

16 UTSA (19-13)	1 Ohio St. (32-2)
16 Ala. St. (17-17)	8 George Mason (26-6)
12 UAB (22-8)	9 Villanova (21-11)
12 Clemson (21-11)	5 West Virginia (20-11)
	4 Kentucky (25-8)
	13 Princeton (25-6)
	6 Xavier (24-7)
	11 Marquette (20-14)
	3 Syracuse (26-7)
	14 Indiana St. (20-13)
	7 Washington (23-10)
	10 Georgia (21-11)
	2 North Carolina (26-7)
	15 Long Island (27-5)

EAST
NEWARK
Mar 25 & 27

1 Duke (30-4)	16 Hampton (24-8)
8 Michigan (20-13)	9 Tennessee (19-14)
5 Arizona (27-7)	12 Memphis (25-9)
4 Texas (27-7)	13 Oakland (25-9)
6 Cincinnati (25-8)	11 Missouri (23-10)
3 Connecticut (26-9)	14 Bucknell (25-8)
7 Temple (25-7)	10 Penn St. (19-14)
2 San Diego St. (32-2)	15 Northern Colo. (21-10)

WEST
ANAHEIM
Mar 24 & 26Final Four®
HOUSTON
APRIL 2-4National
Championship
APRIL 4Watch the tournament on these networks
or online at MHOD.NCAA.COM

First Four

Dayton

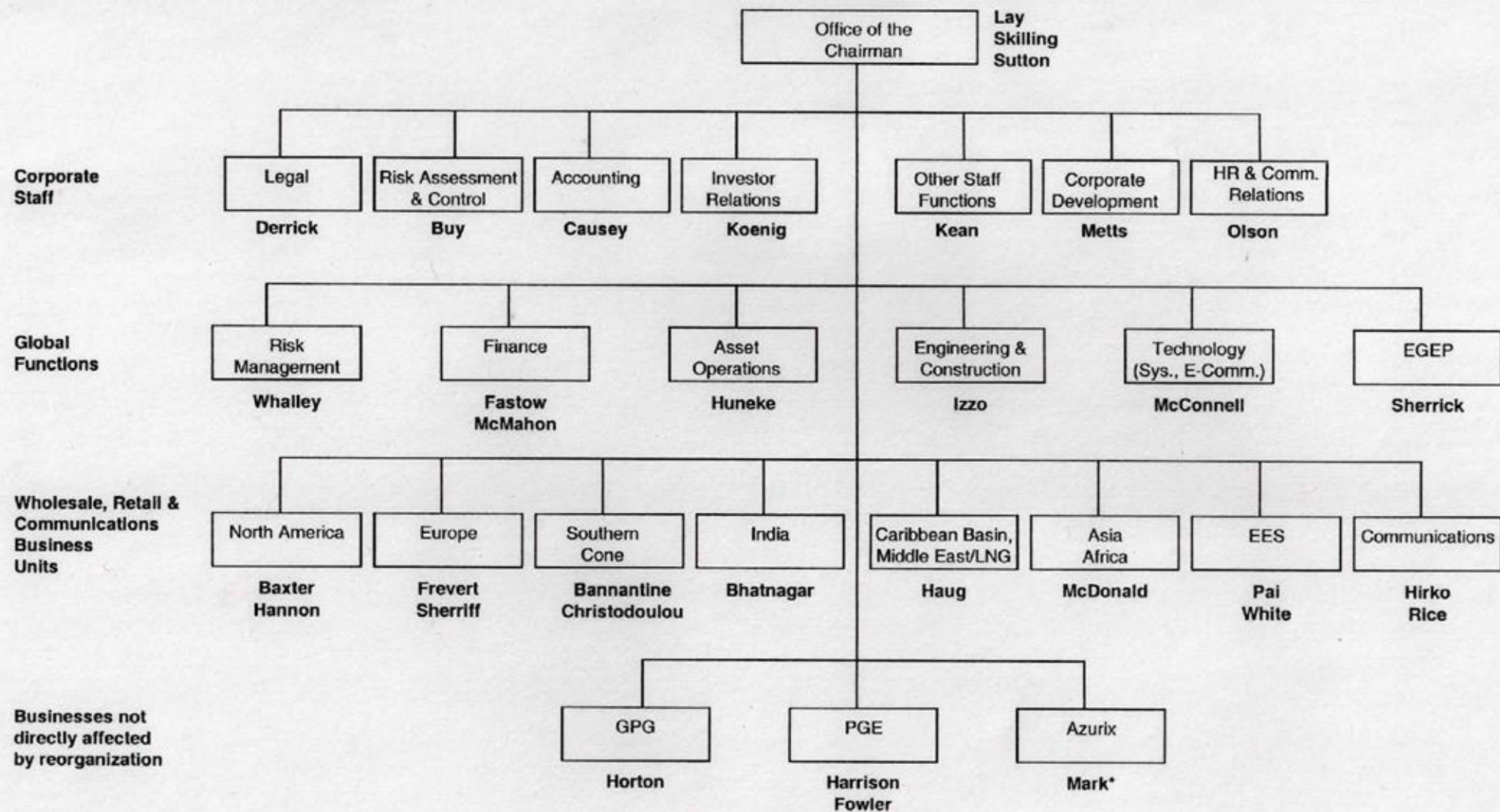
1 Kansas (32-2)	16 Boston U. (21-13)
8 UNLV (24-8)	9 Illinois (19-13)
5 Vanderbilt (23-10)	12 Richmond (27-7)
4 Louisville (25-9)	13 Morehead St. (24-9)
6 Georgetown (21-10)	11 USC (19-14)
	11 VCU (23-8)
	3 Purdue (25-7)
	14 St. Peter's (20-13)
	7 Texas A&M (24-8)
	10 Florida St. (21-10)
	2 Notre Dame (26-6)
	15 Akron (23-12)

SOUTHWEST
SAN ANTONIO
Mar 25 & 27

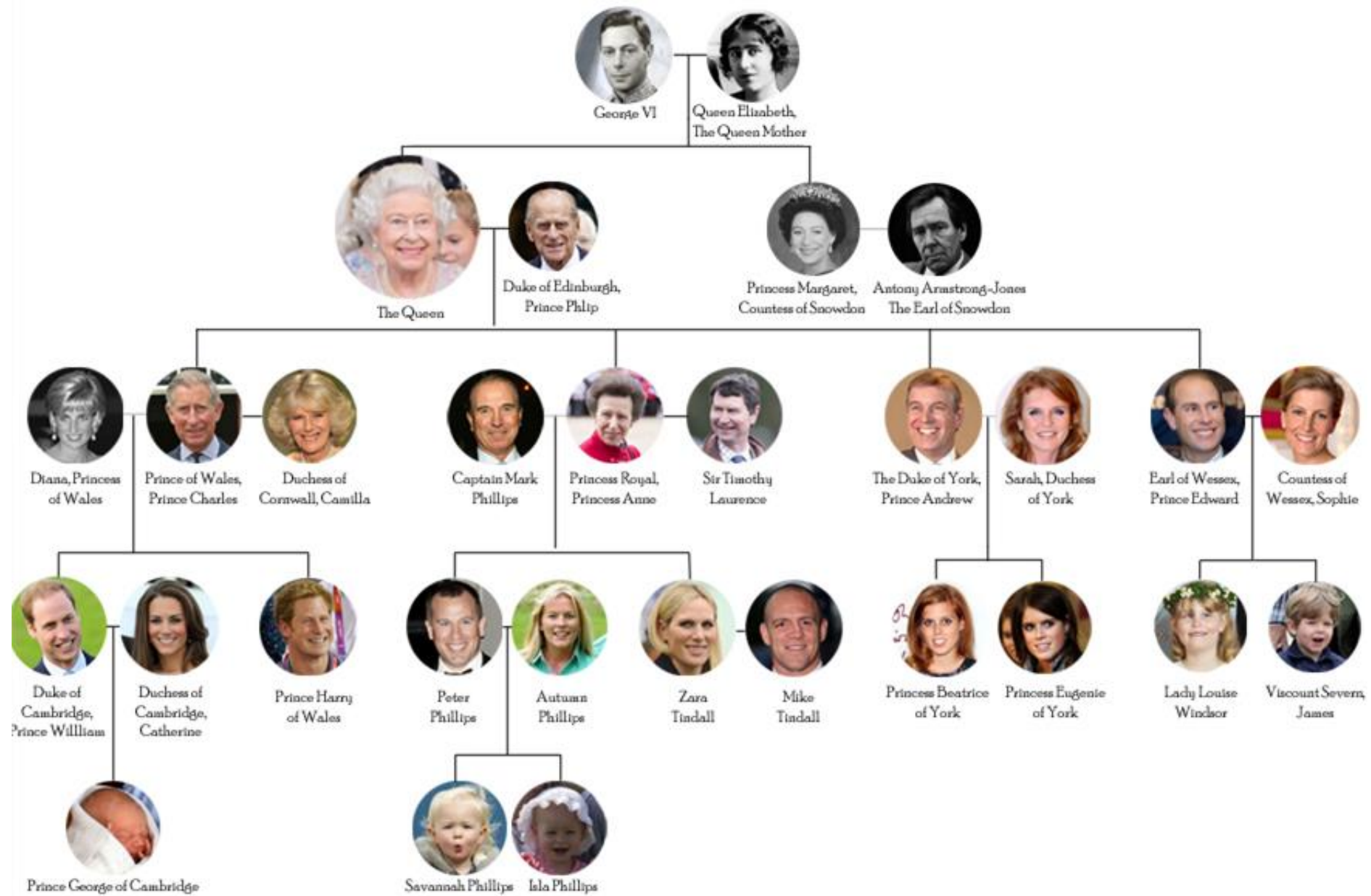
1 Pittsburgh (27-5)	16 UNC Ash. (19-13)
8 Butler (23-9)	16 UALR (19-16)
9 Old Dominion (27-6)	
5 Kansas St. (22-10)	
12 Utah St. (30-3)	
4 Wisconsin (23-8)	
13 Belmont (30-4)	
6 St. John's (NY) (21-11)	
11 Gonzaga (24-9)	
3 BYU (30-4)	
14 Wofford (21-12)	
7 UCLA (22-10)	
10 Michigan St. (19-14)	
2 Florida (26-7)	
15 UC Santa Barbara (18-13)	

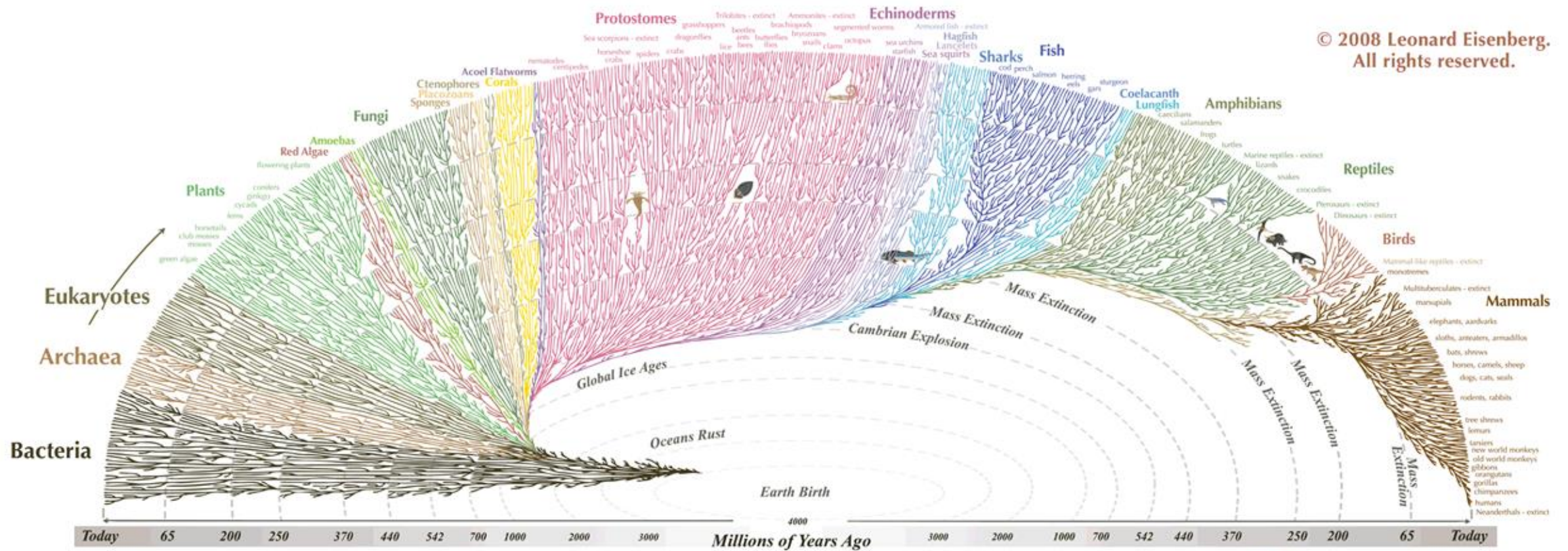
SOUTHEAST
NEW ORLEANS
Mar 24 & 26

ENRON CORPORATE ORGANIZATION AND EXECUTIVE COMMITTEE MEMBERS



*Ad hoc member

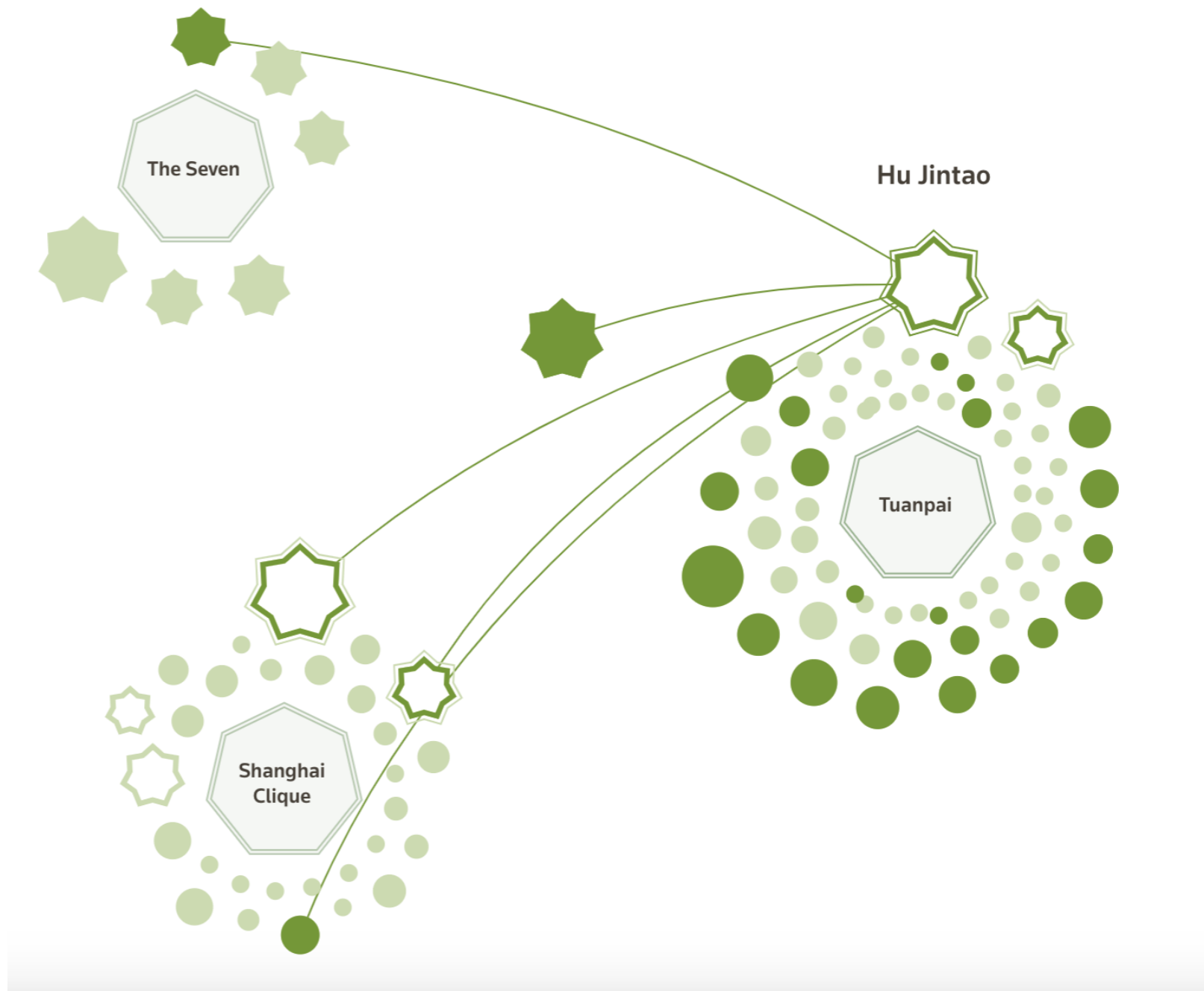




All the major and many of the minor living branches of life are shown on this diagram, but only a few of those that have gone extinct are shown. Example: Dinosaurs - extinct



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evogeneo.com



Politburo



Current



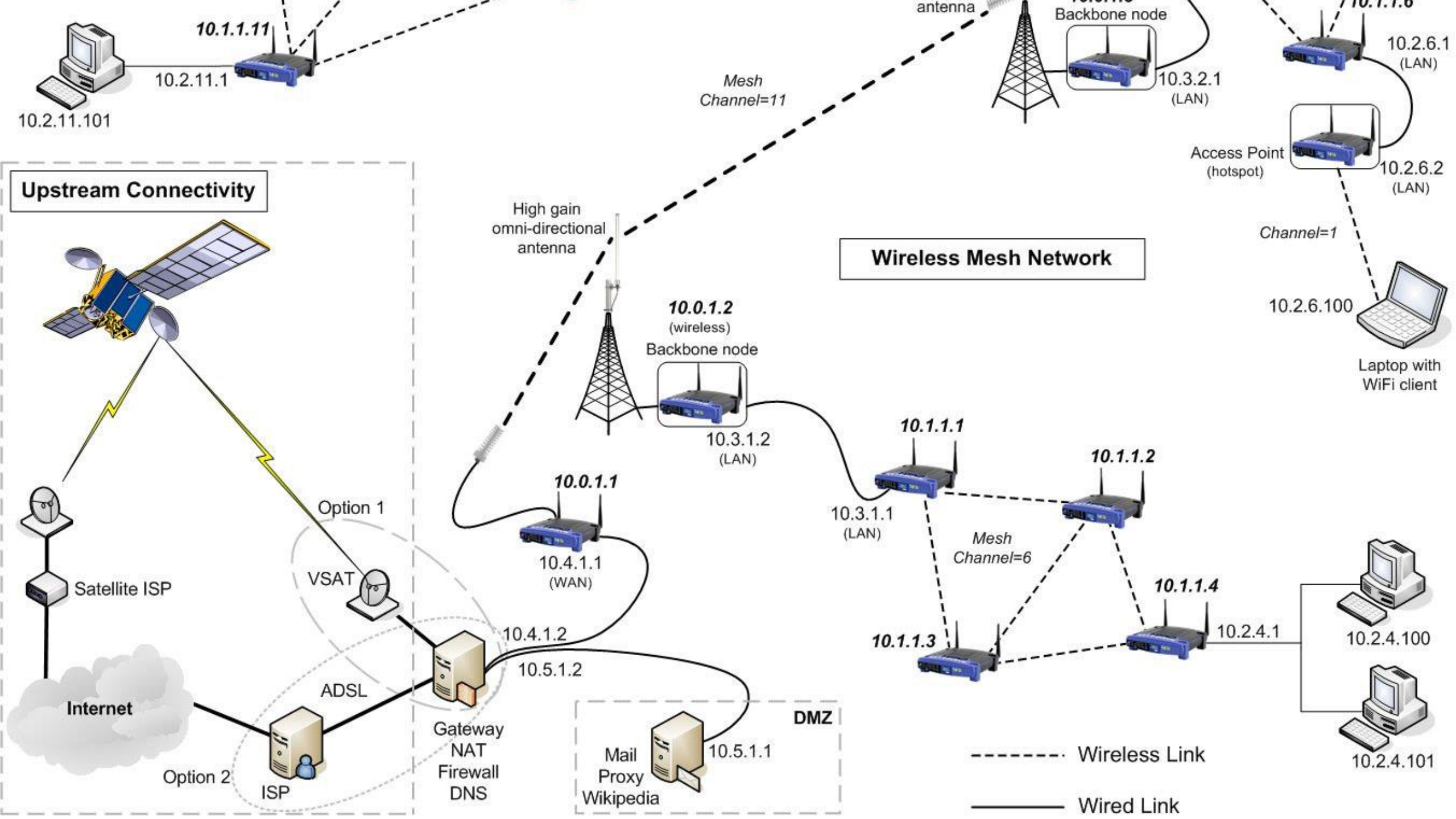
Former

Leaders



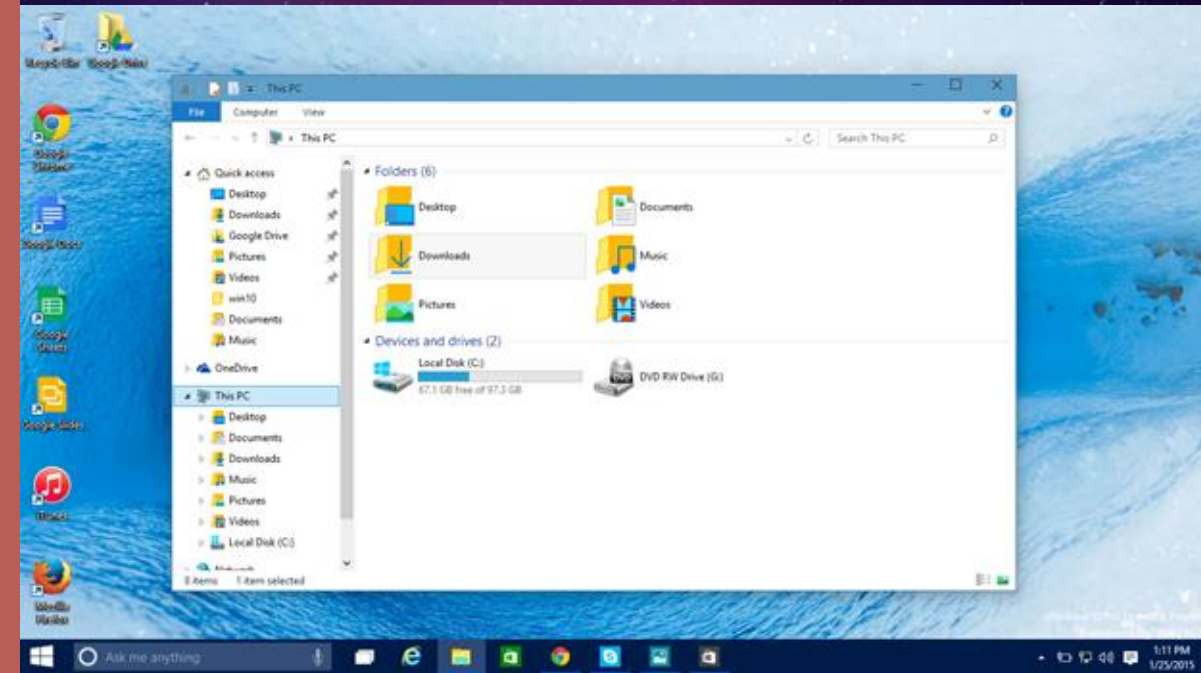
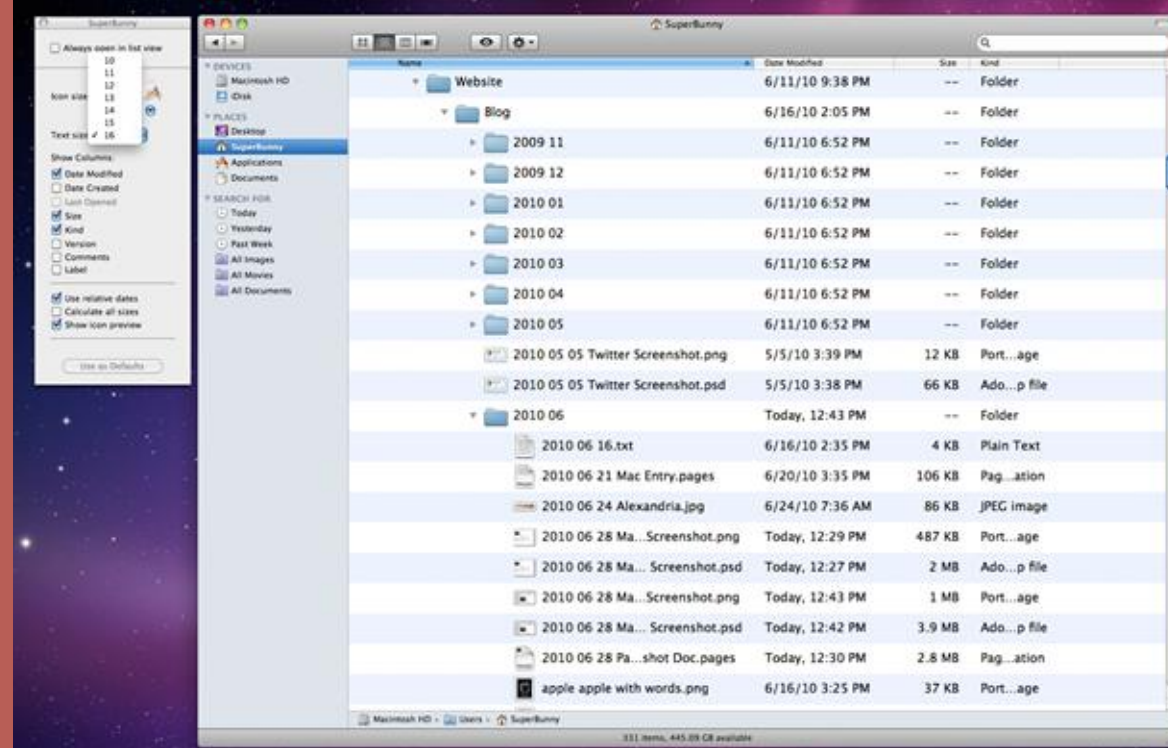
Importance





Need to navigate
through the **file**
hierarchy to find a
particular file.

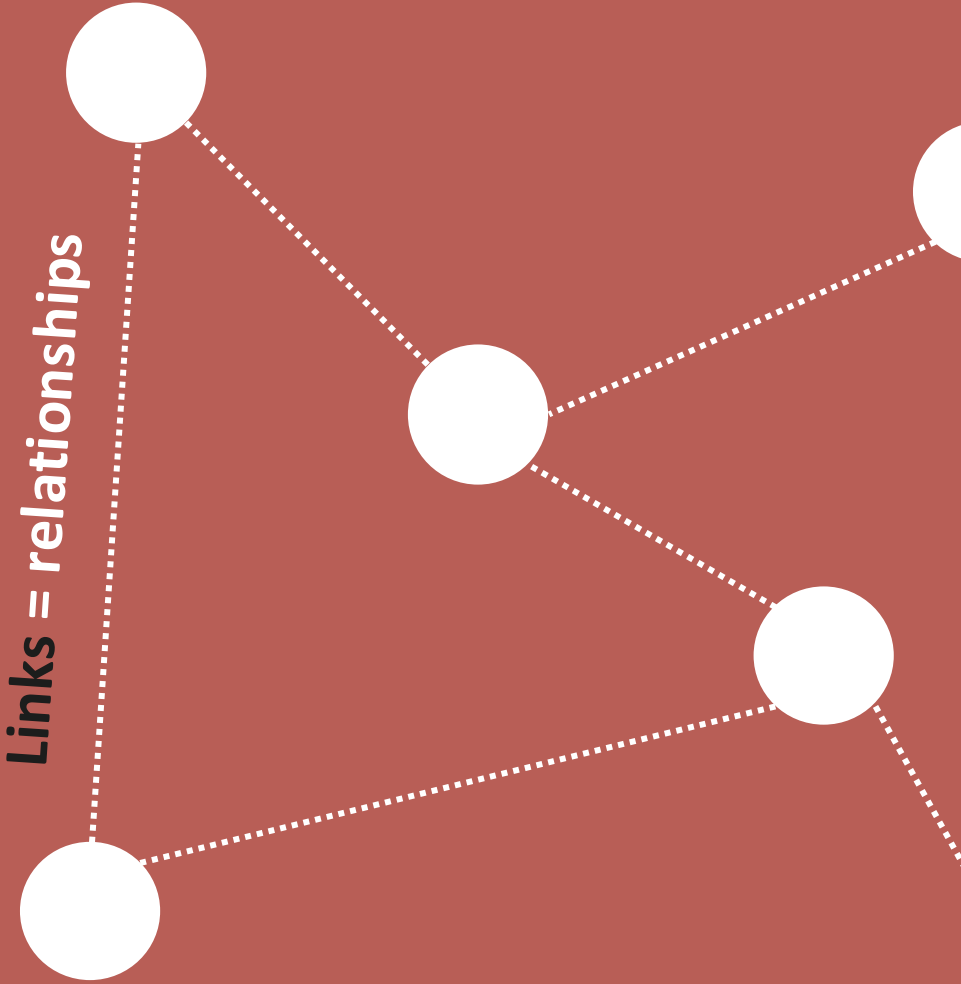
Where am I?
What am I looking for?

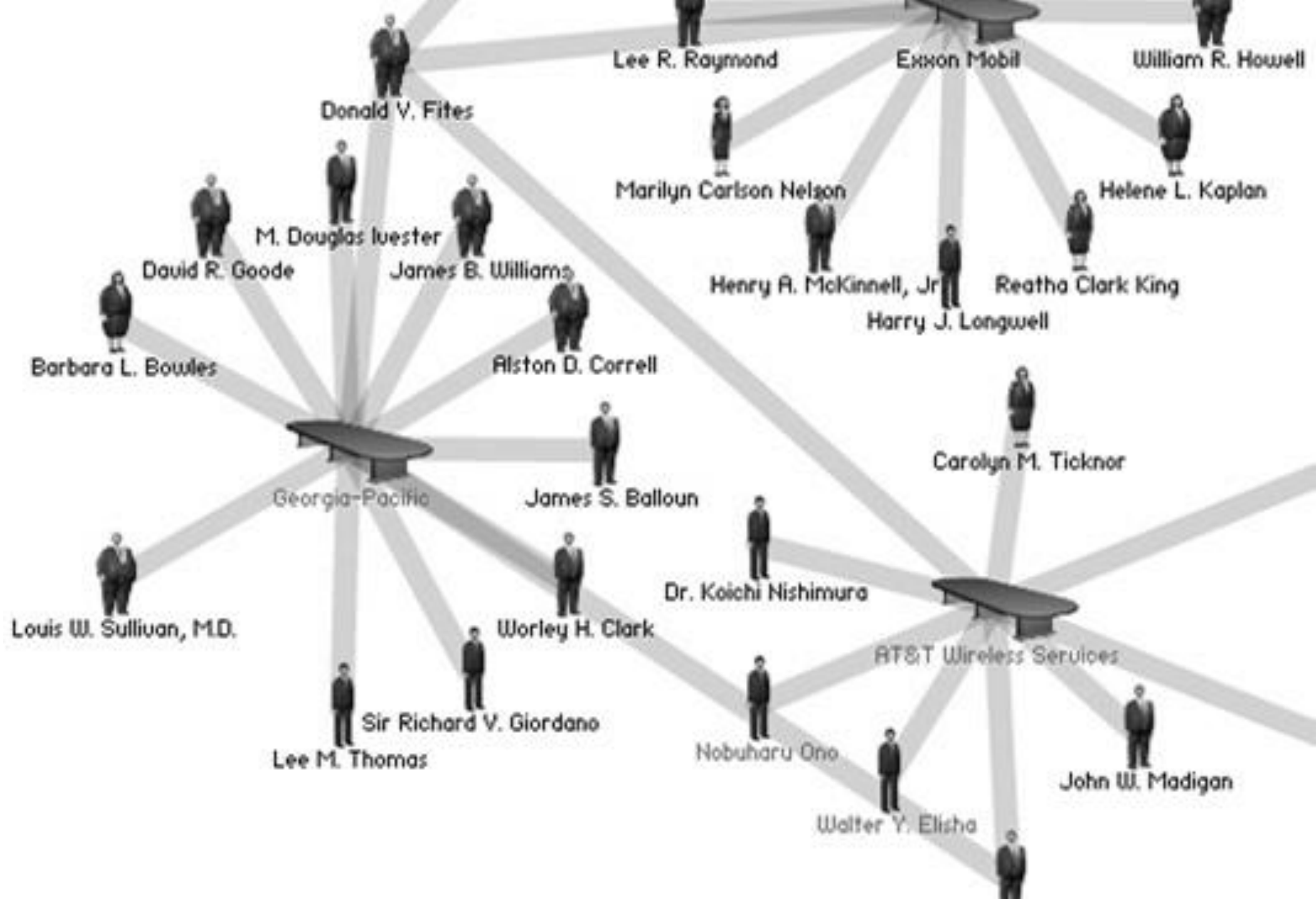


Representations of structured, connected data

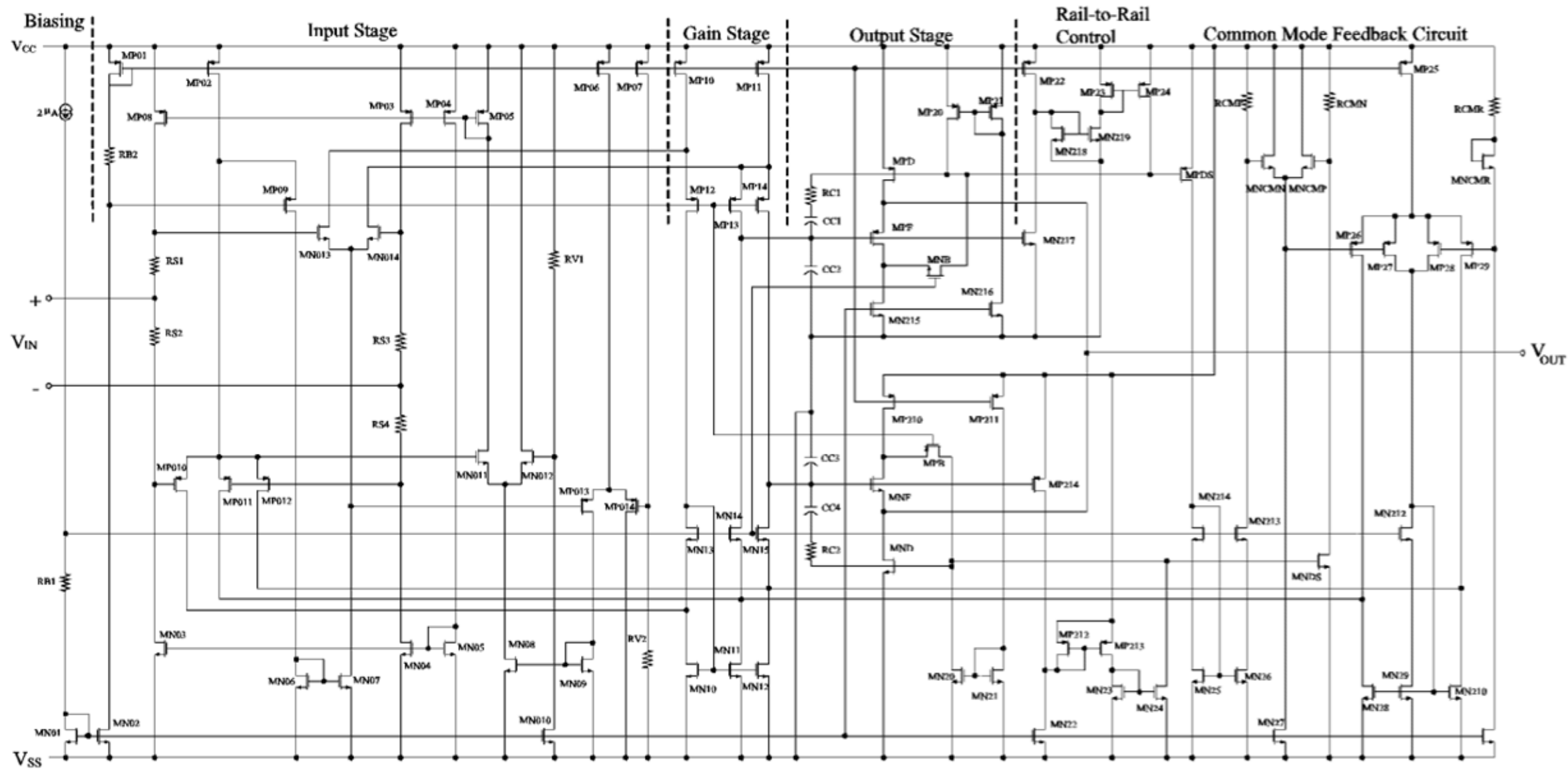
Nodes
= data

Links = relationships





They Rule



Integrated Circuit



Relationships of Java Classes

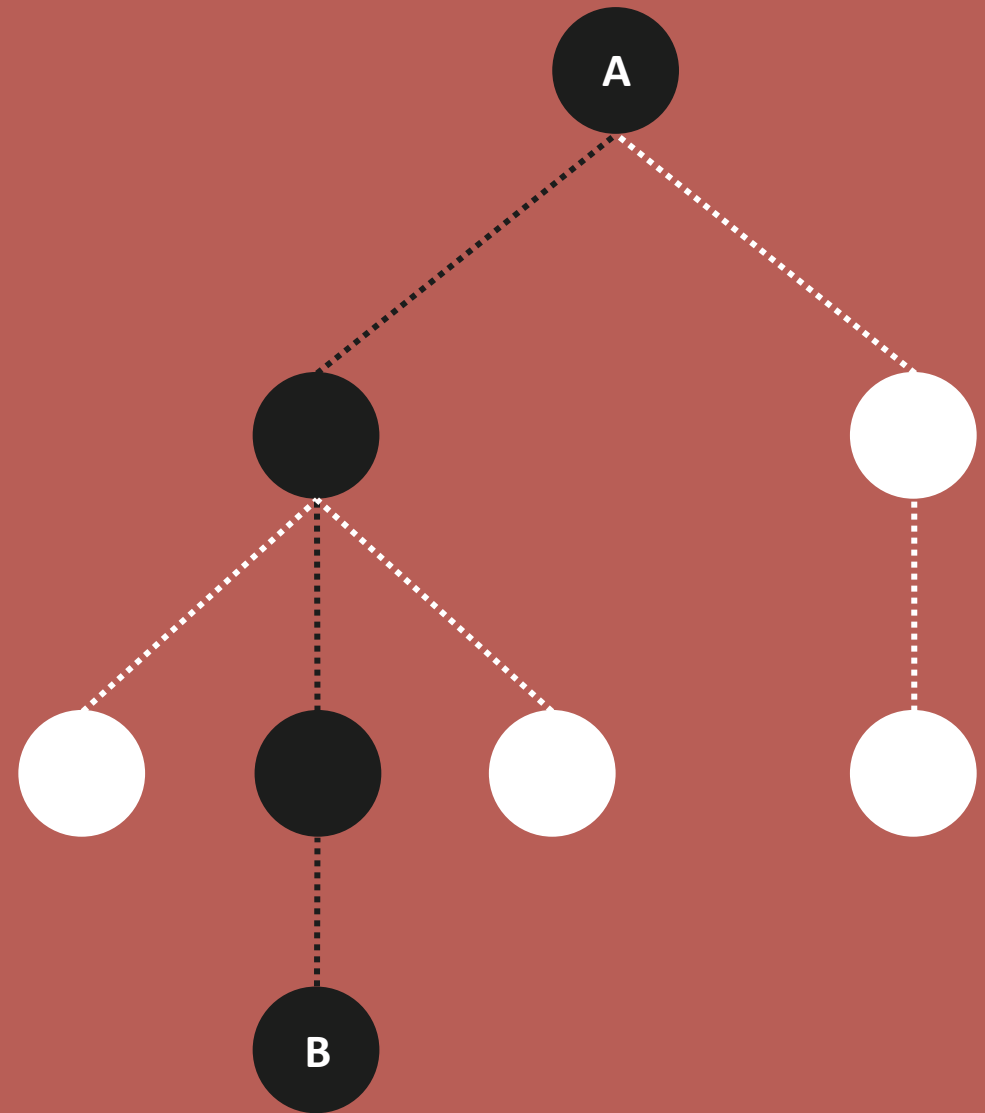
What?

Connectivity/path-following

Network distance

Clustering

Ordering/hierarchy level

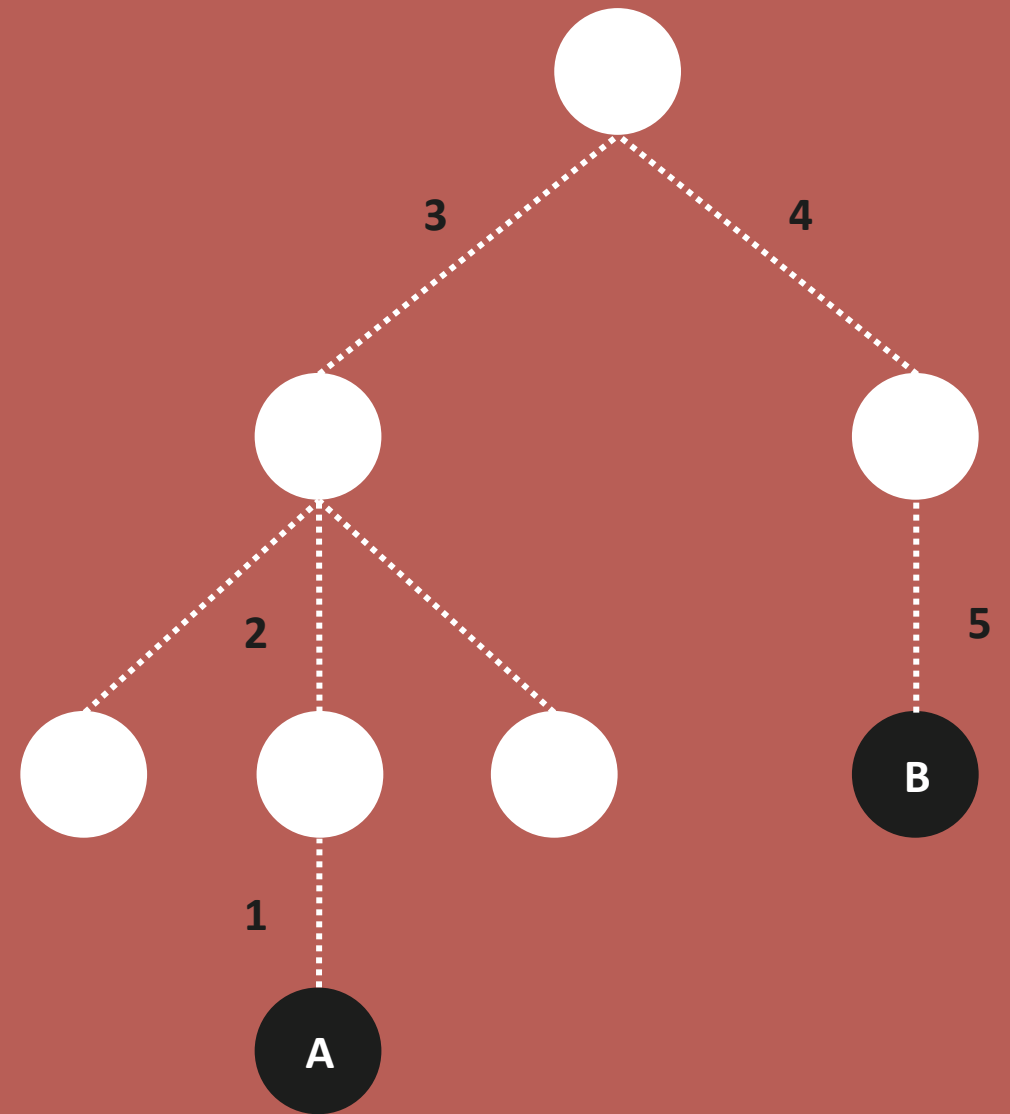


Connectivity/path-following

Network distance

Clustering

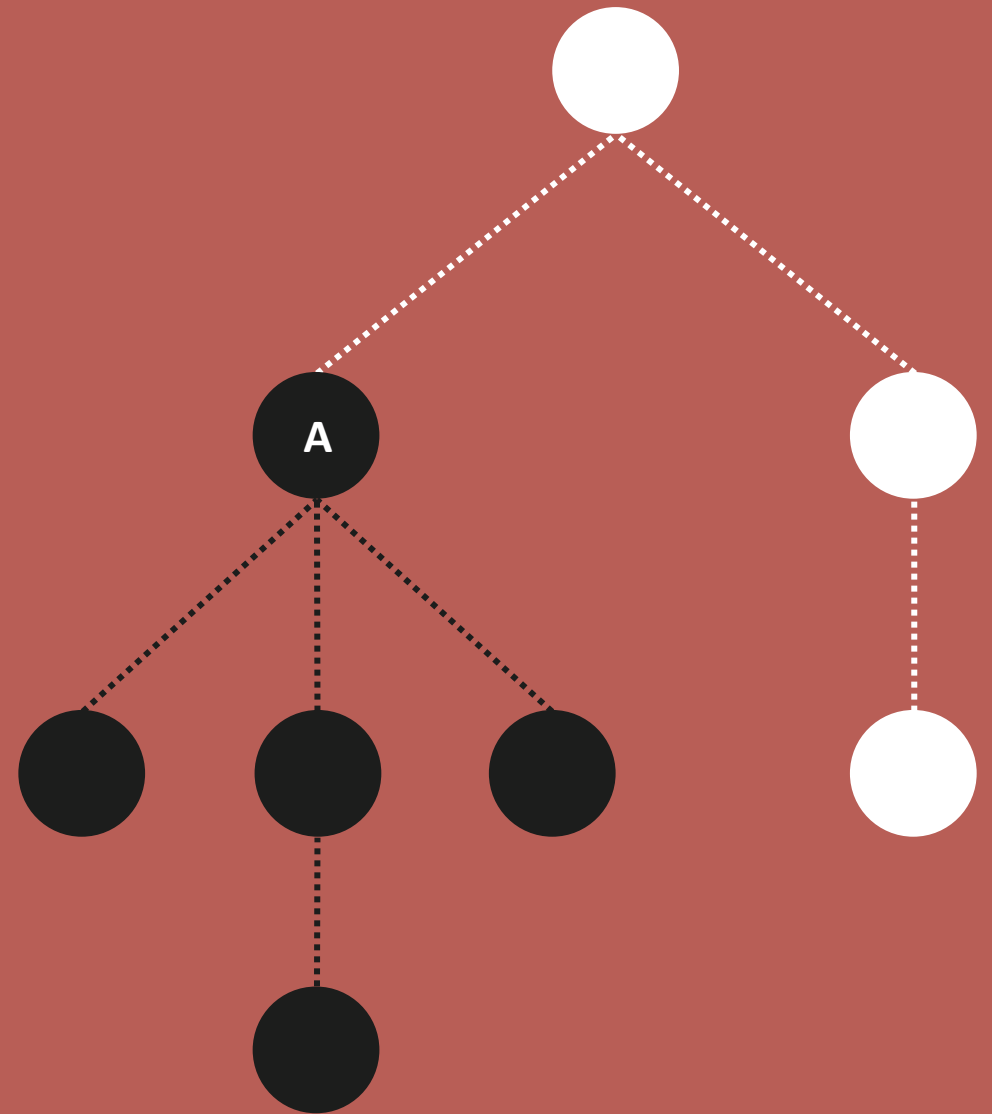
Ordering/hierarchy level



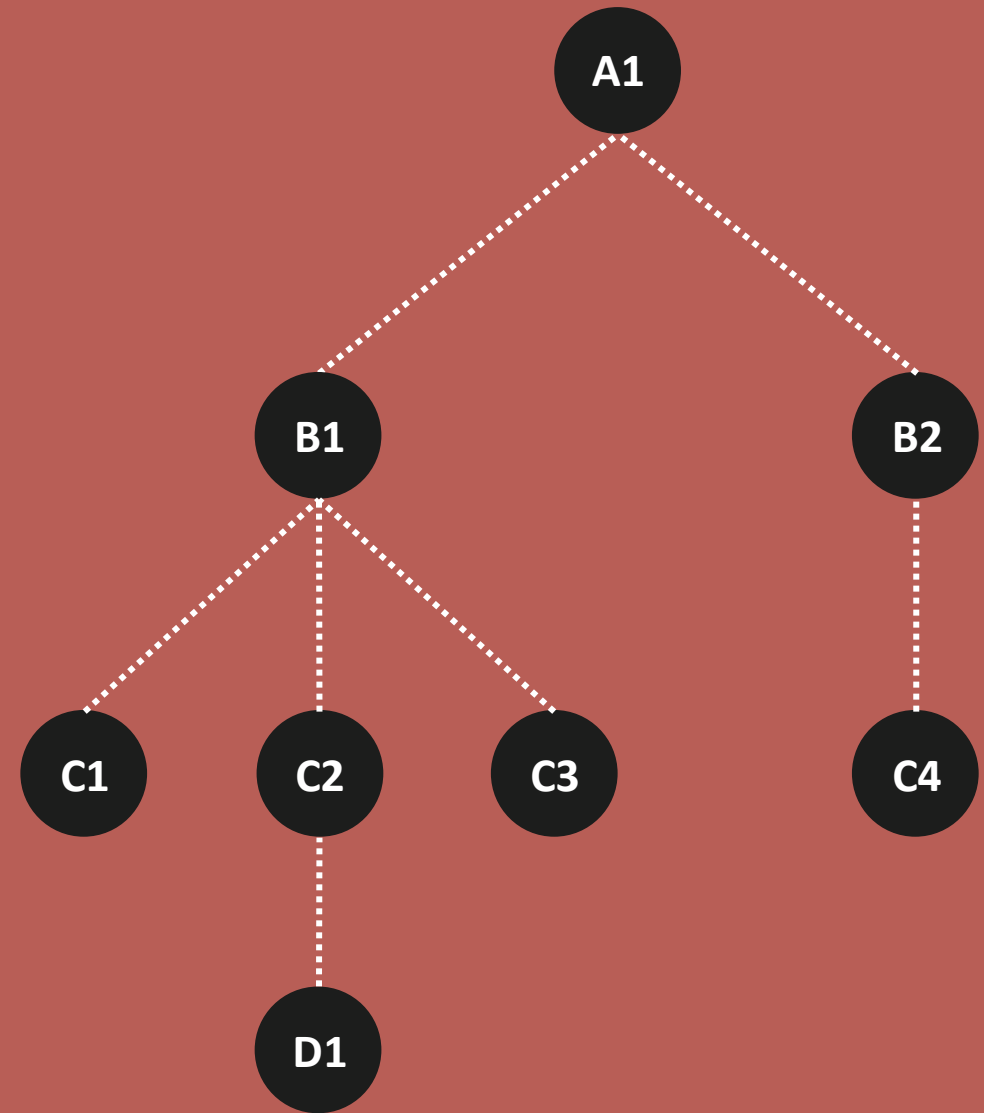
Connectivity/path-following
Network distance

Clustering

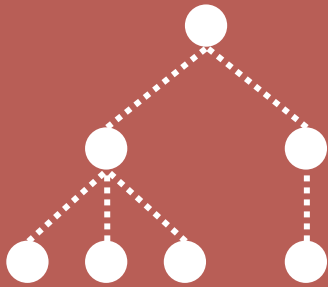
Ordering/hierarchy level



Connectivity/path-following
Network distance
Clustering
Ordering/hierarchy level



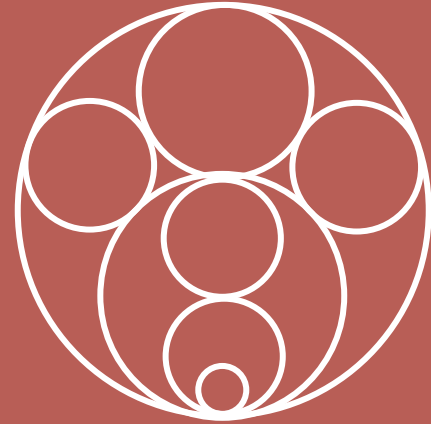
How?



Node-Link Diagrams



Indentation



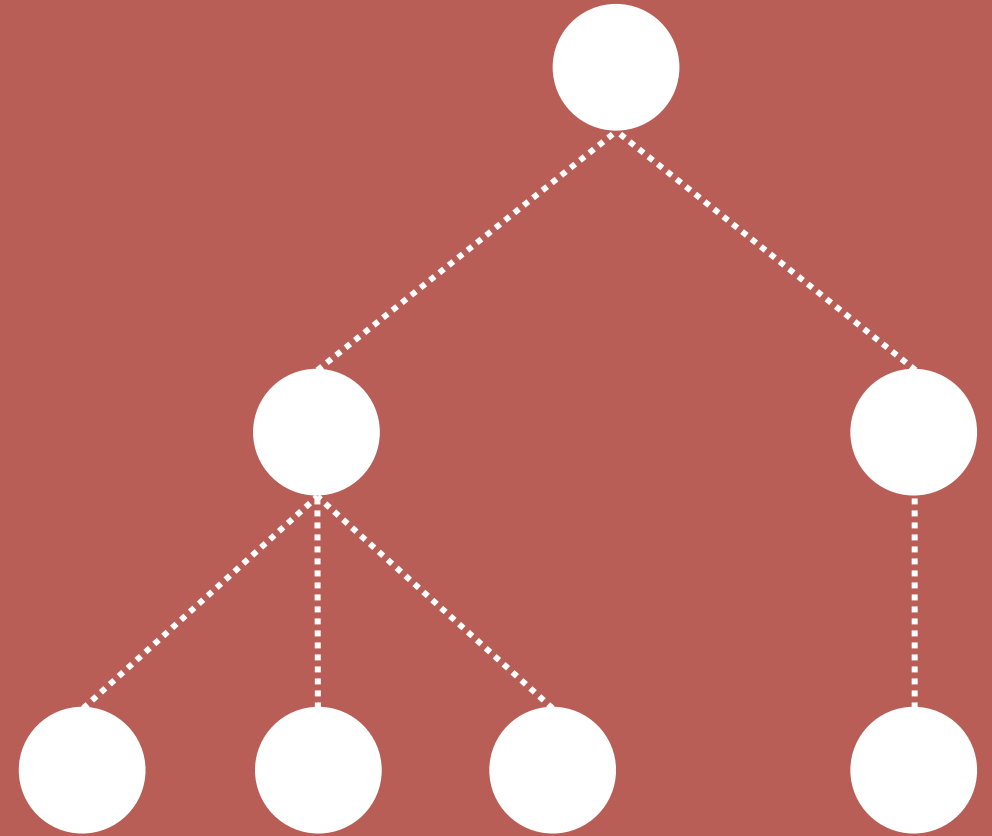
Enclosure

Top-down layout

**Places all items along
vertically spaced rows**

**Uses indentation to show
parent child relationships**

**Breadth and depth end up
fighting for space resources**



Reingold-Tilford Layout

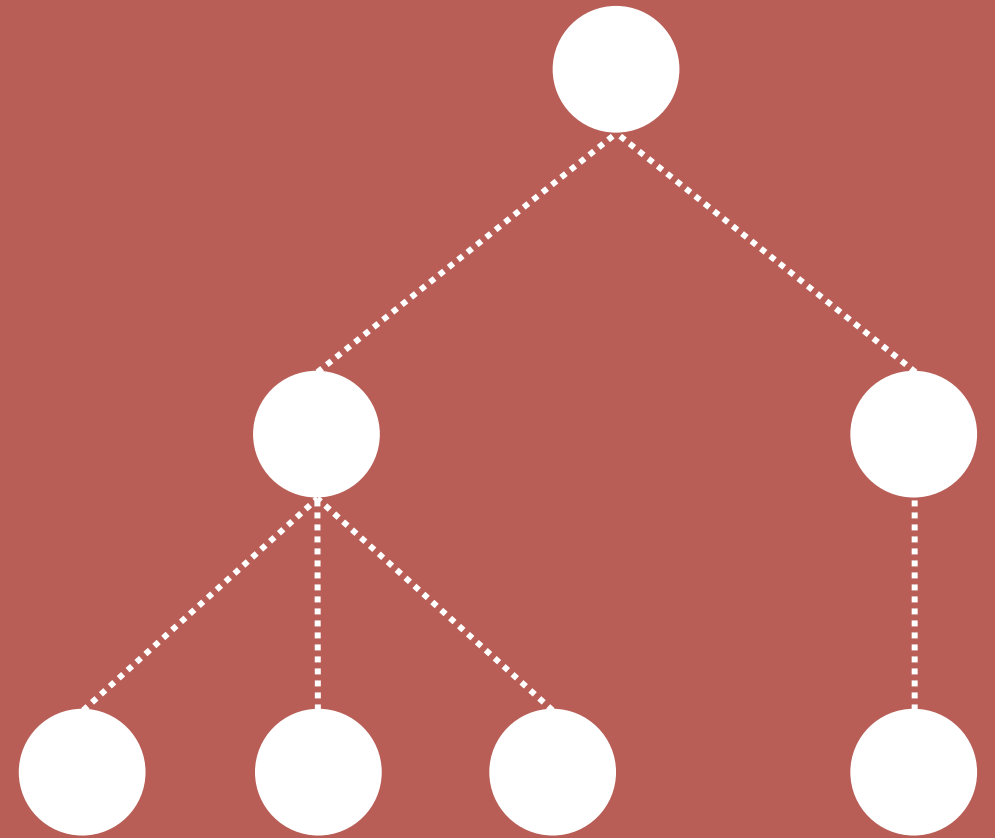
Clearly encodes depth level

No edge crossings

Isomorphic subtrees drawn identically

Ordering and symmetry preserved

Compact layout



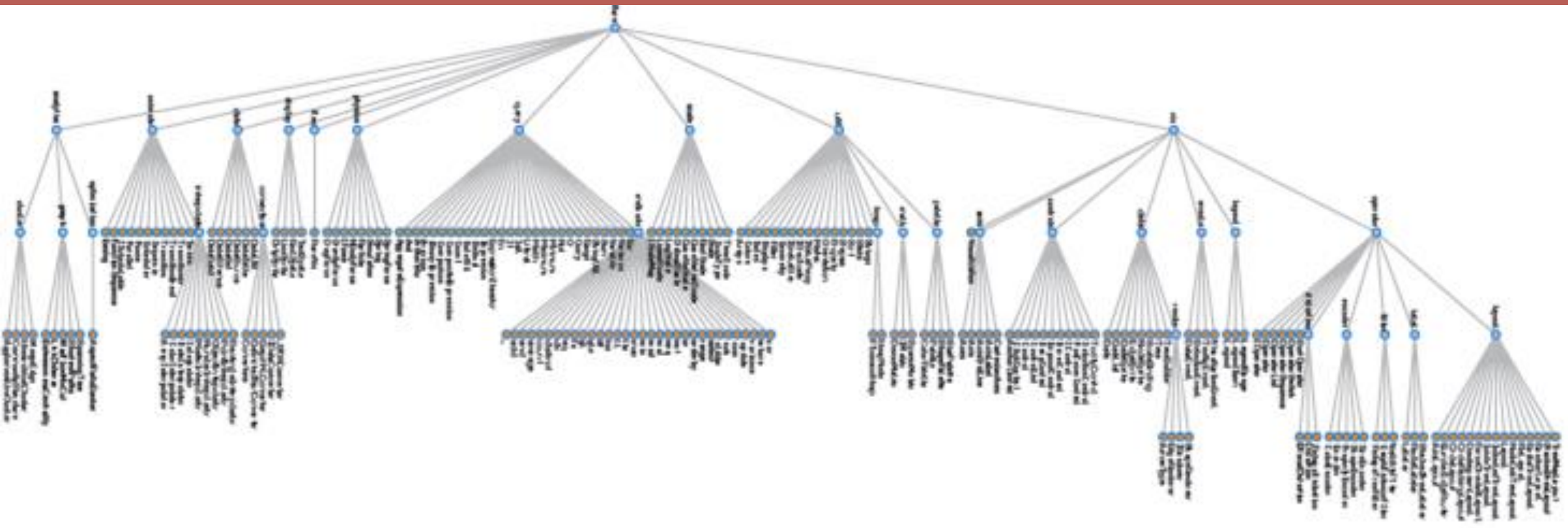
Reingold-Tilford Layout

Scale becomes a problem.

Tree breadth often grows exponentially.
Even with tidier layout, quickly run out of space.

Possible solutions:

Filtering
Focus + Context
Scrolling or Panning
Zooming
Aggregation

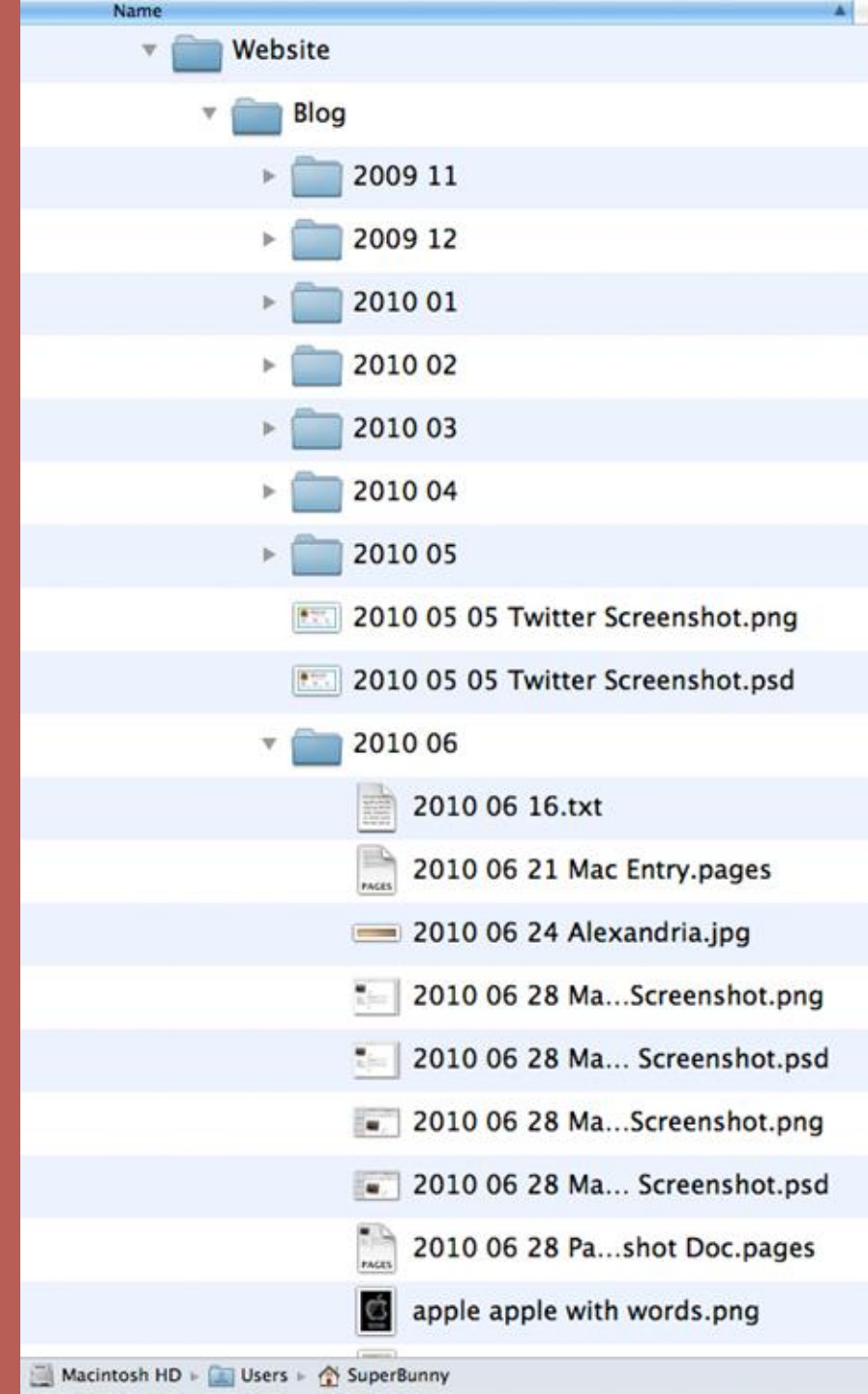


**Places all items along
vertically spaced rows**

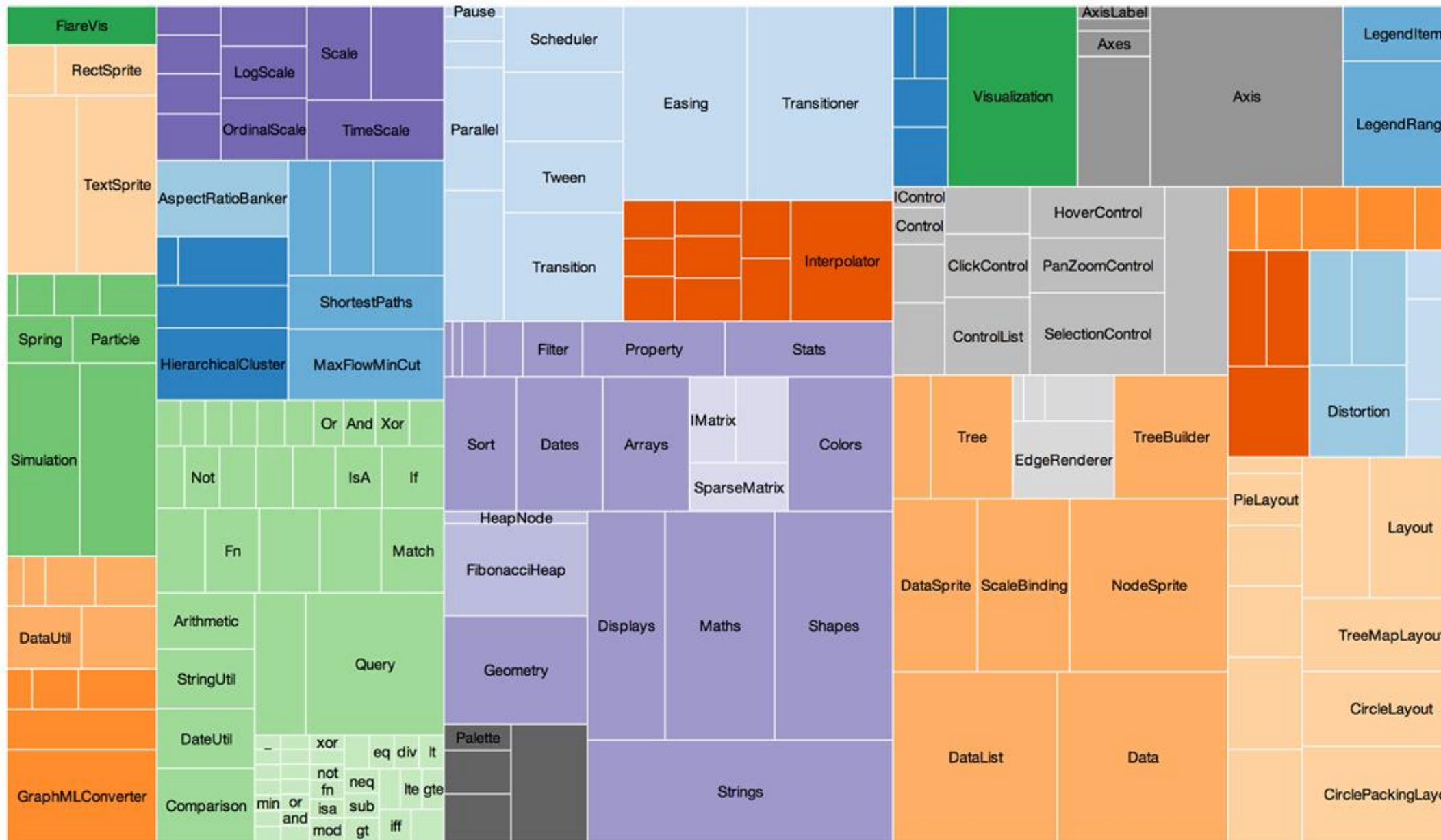
**Uses indentation to show
parent child relationships**

**Breadth and depth end up
fighting for space resources**

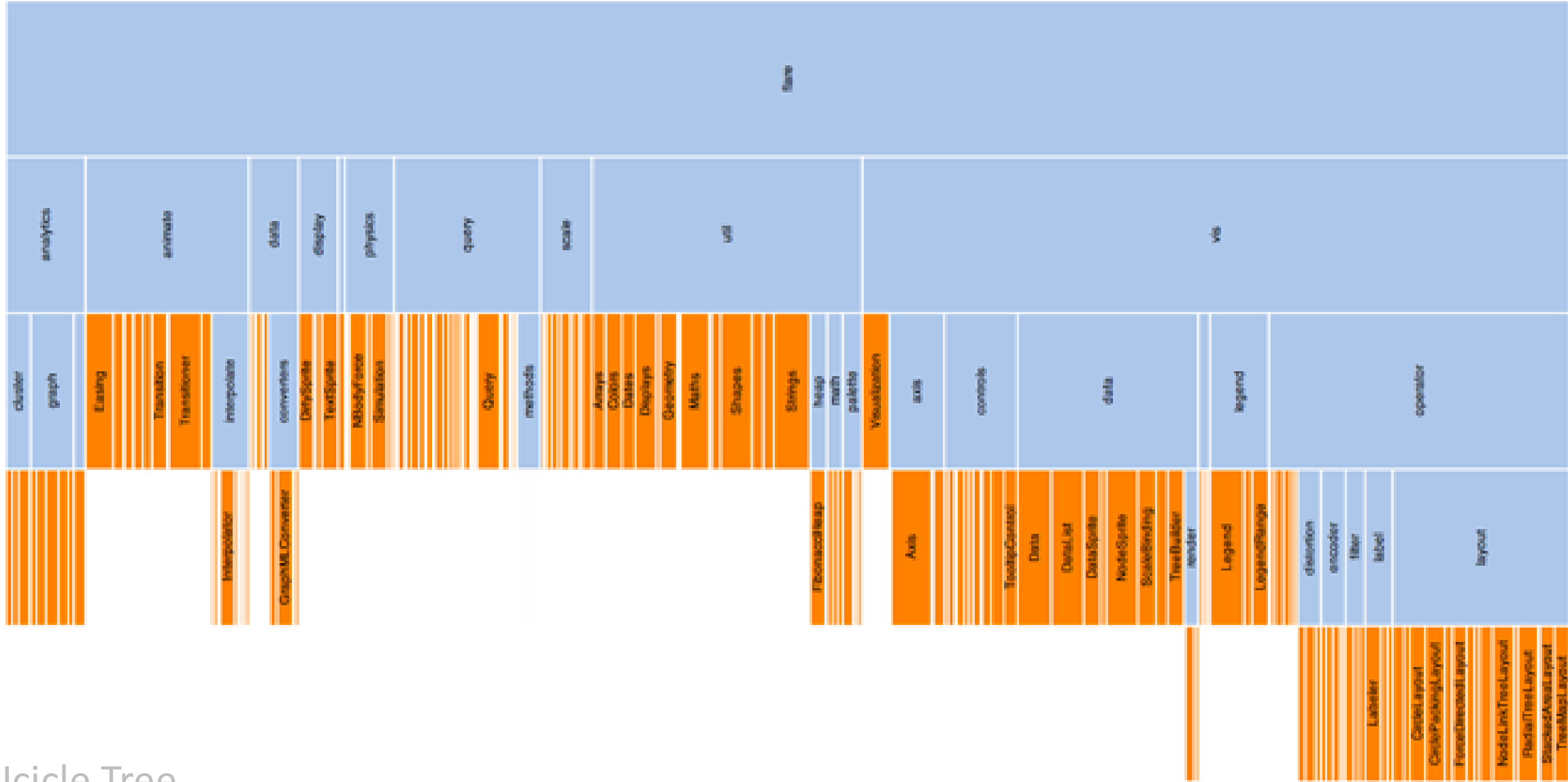
Indented Layout

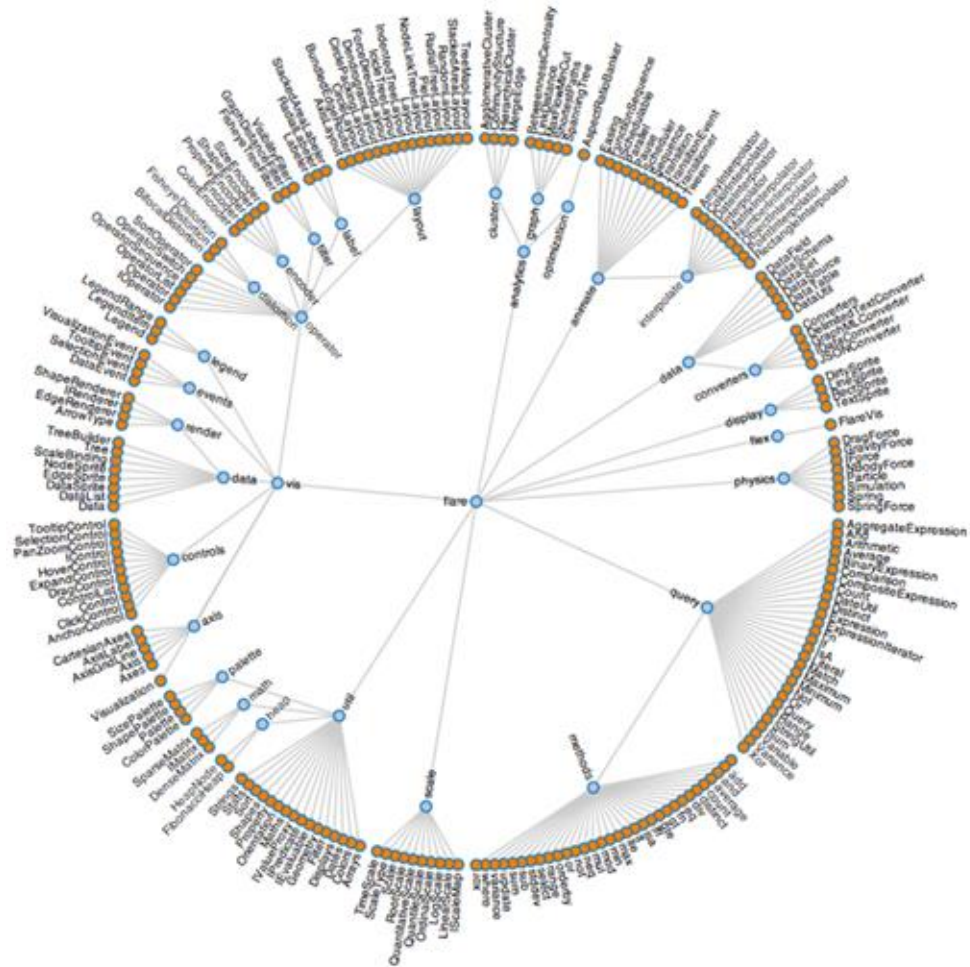


Treemap

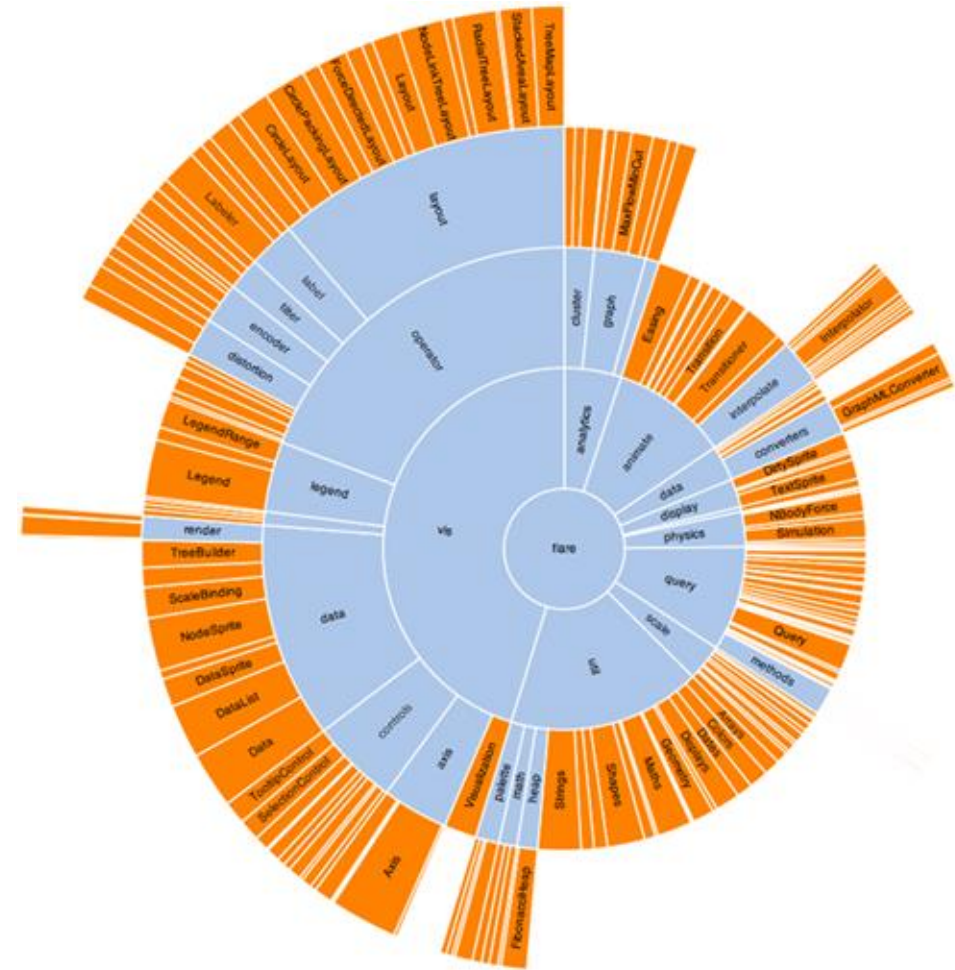


Icicle Tree





Node-link radial layout



Radial icicle

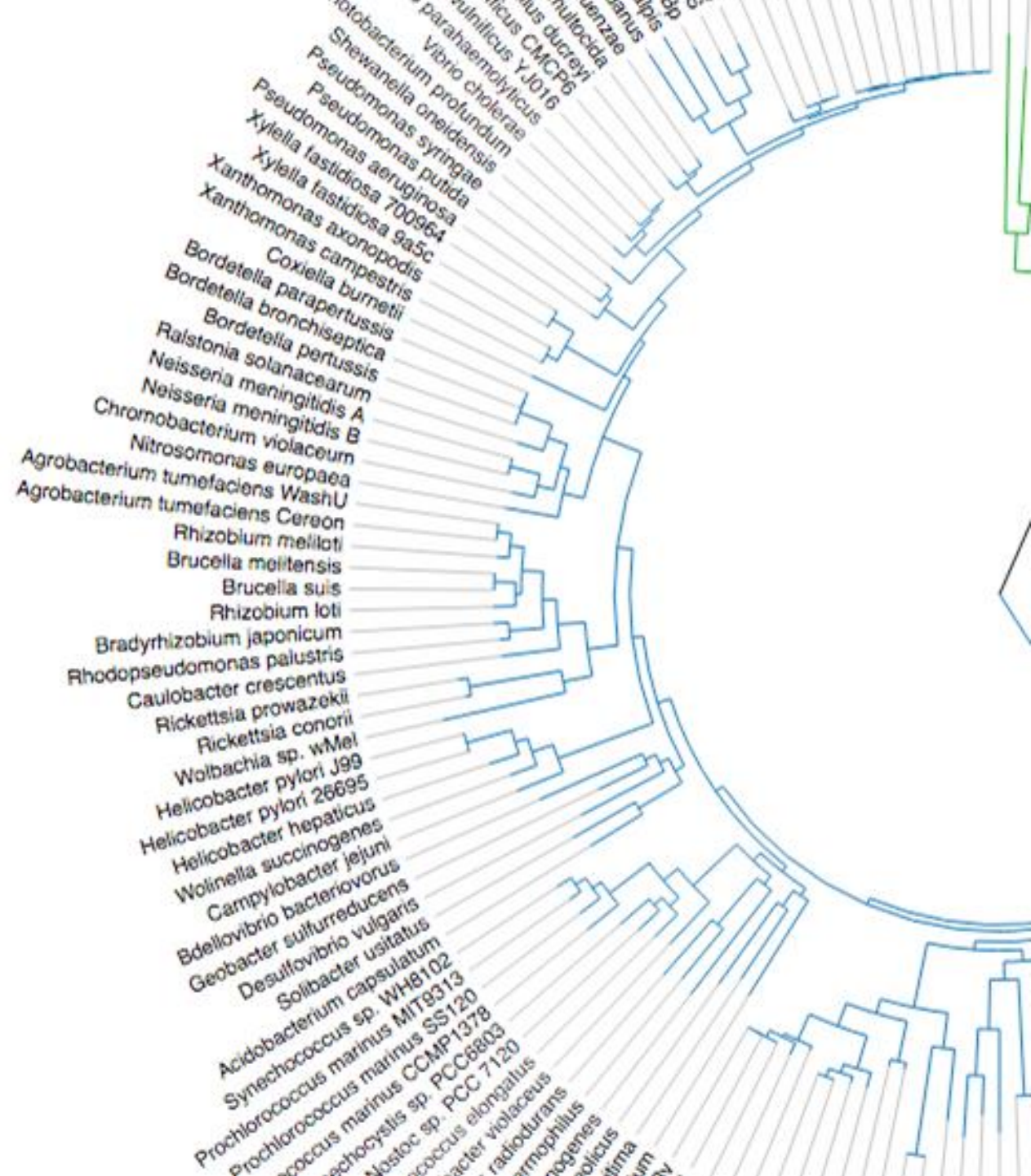
Consider tasks & trade-offs

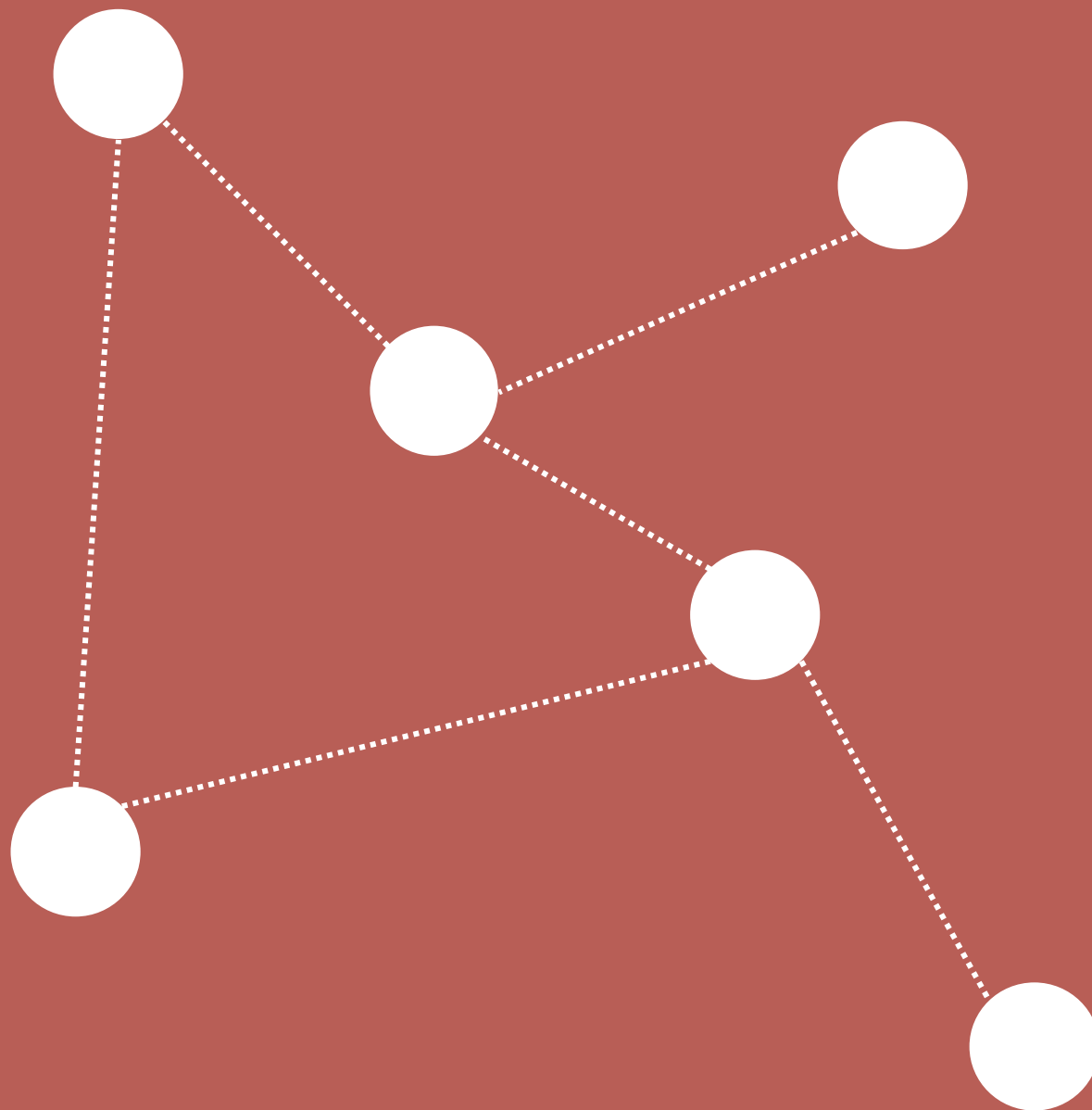
**Does your topic / domain
have established
conventions?**

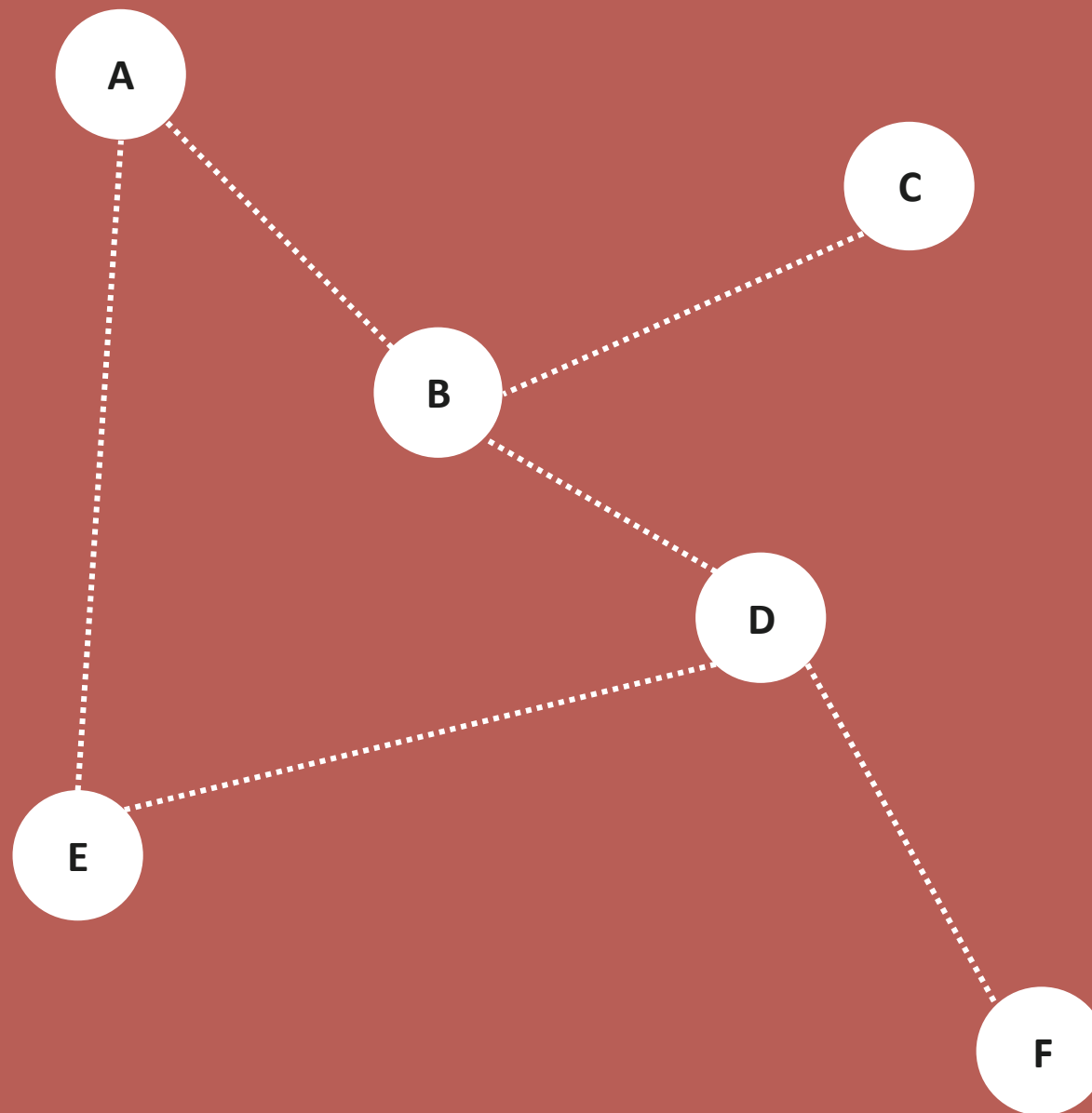
**How much screen real-estate
do you give the tree?**

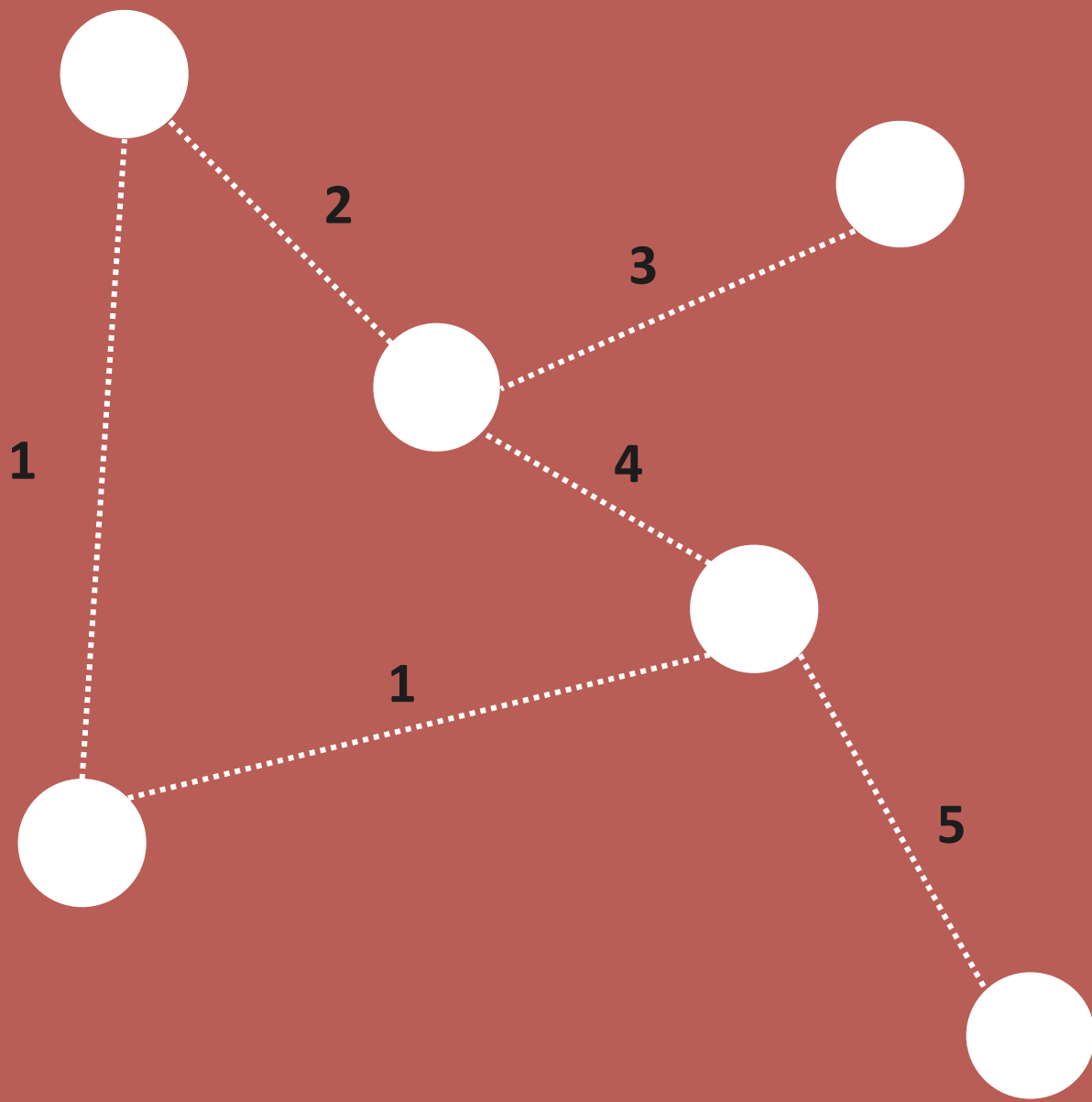
**Designs with multiple
coordinated views are
challenging**

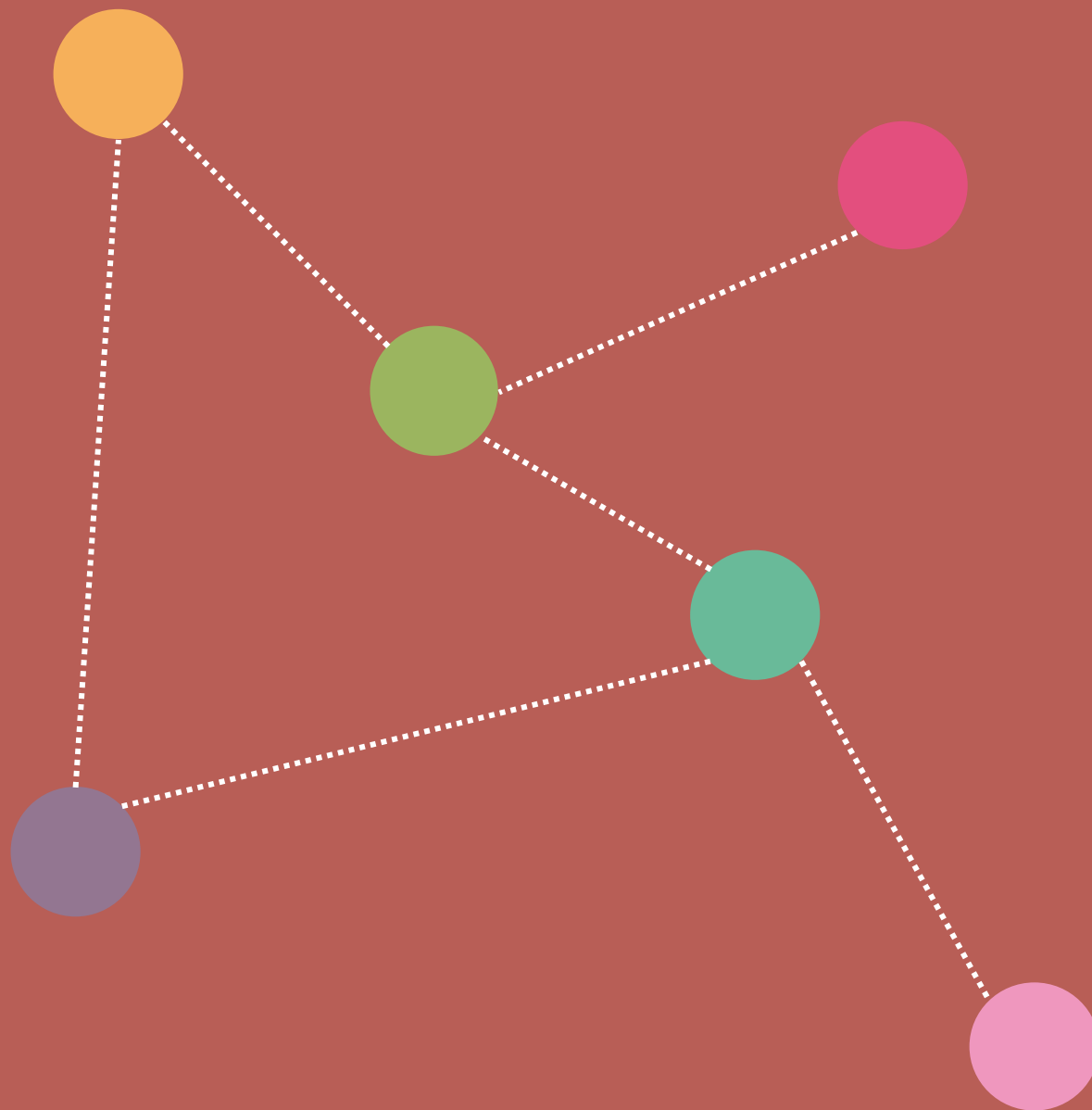
**(Sometimes that indented
tree works the best!)**

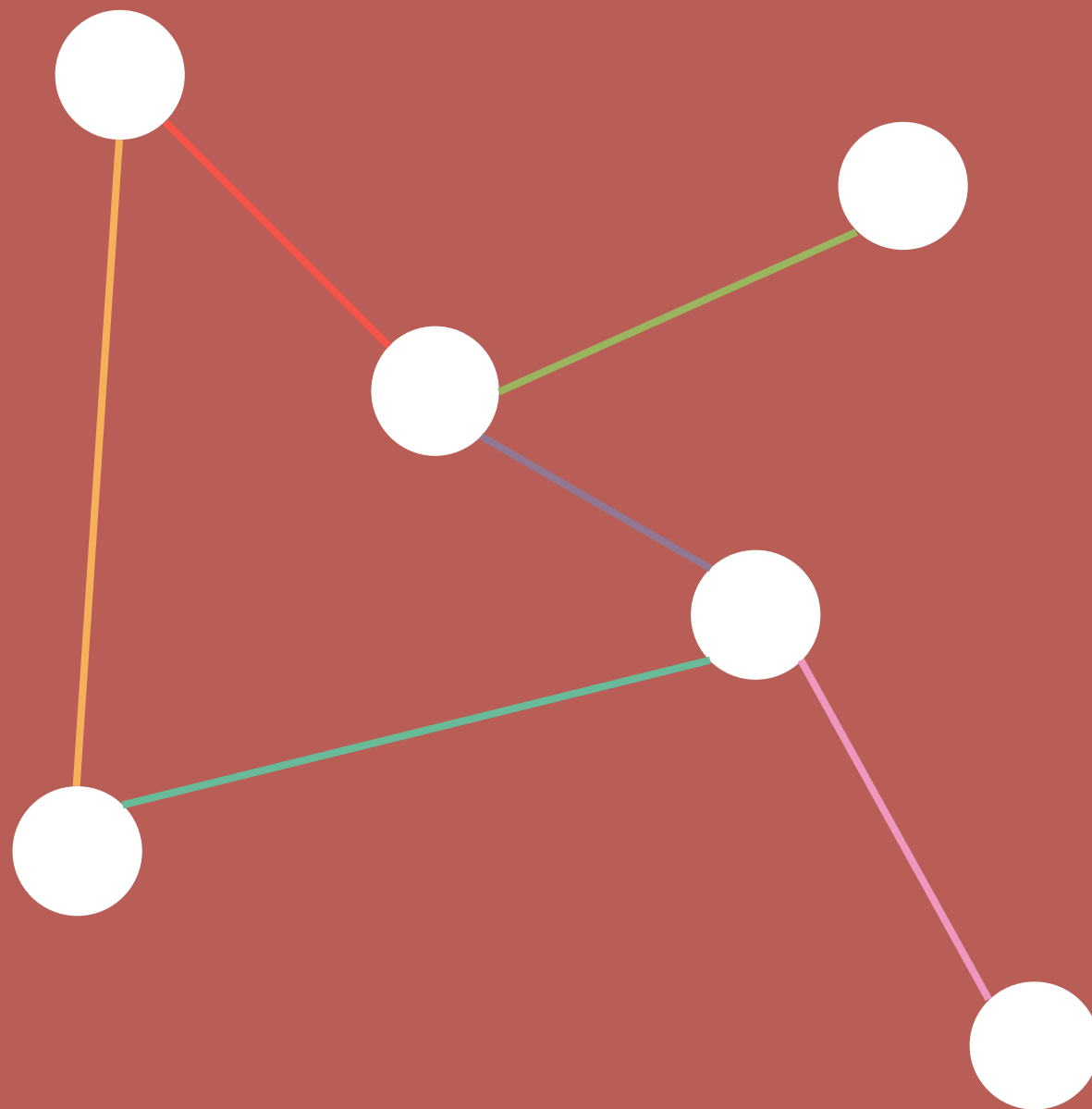


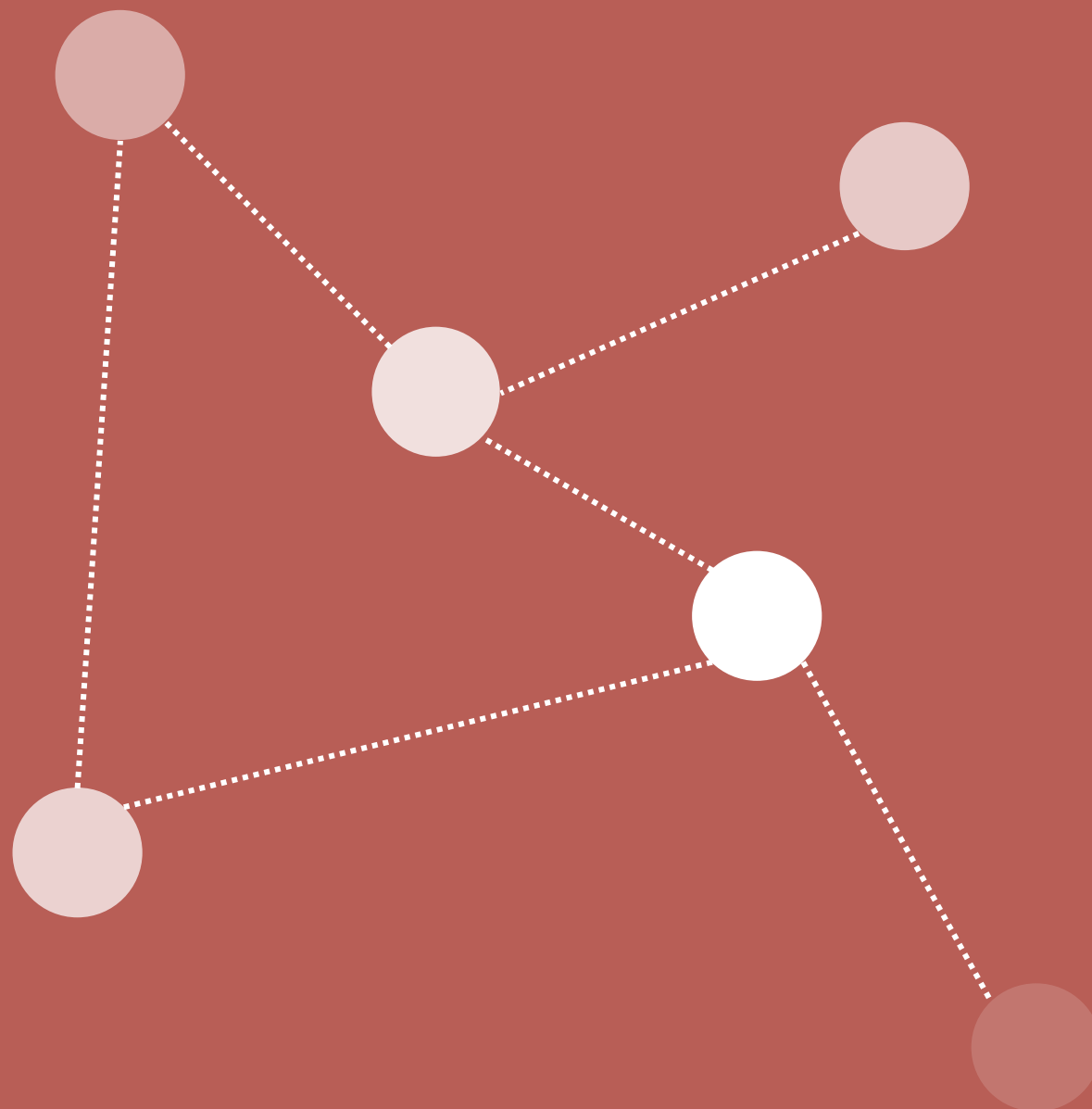


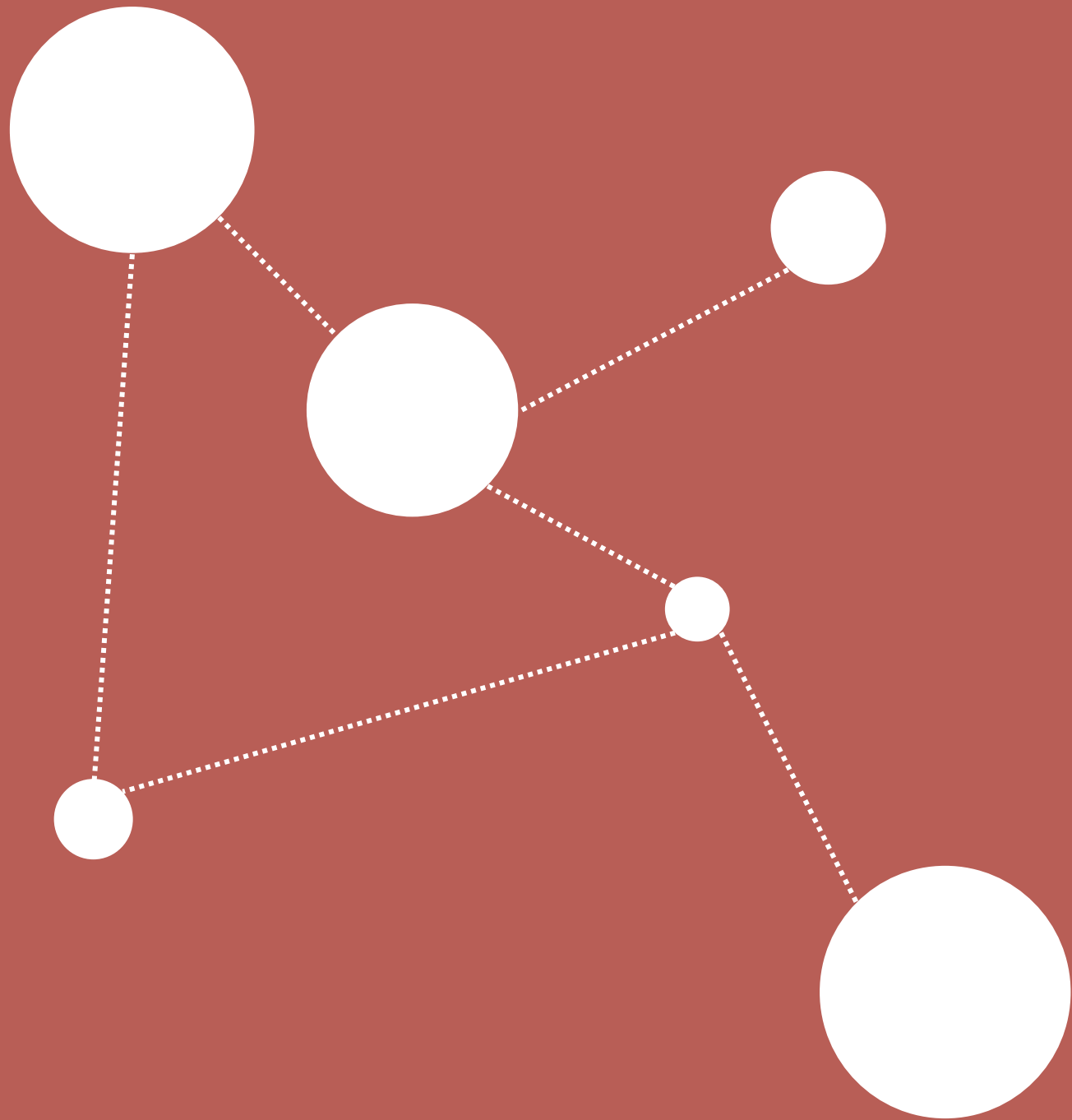


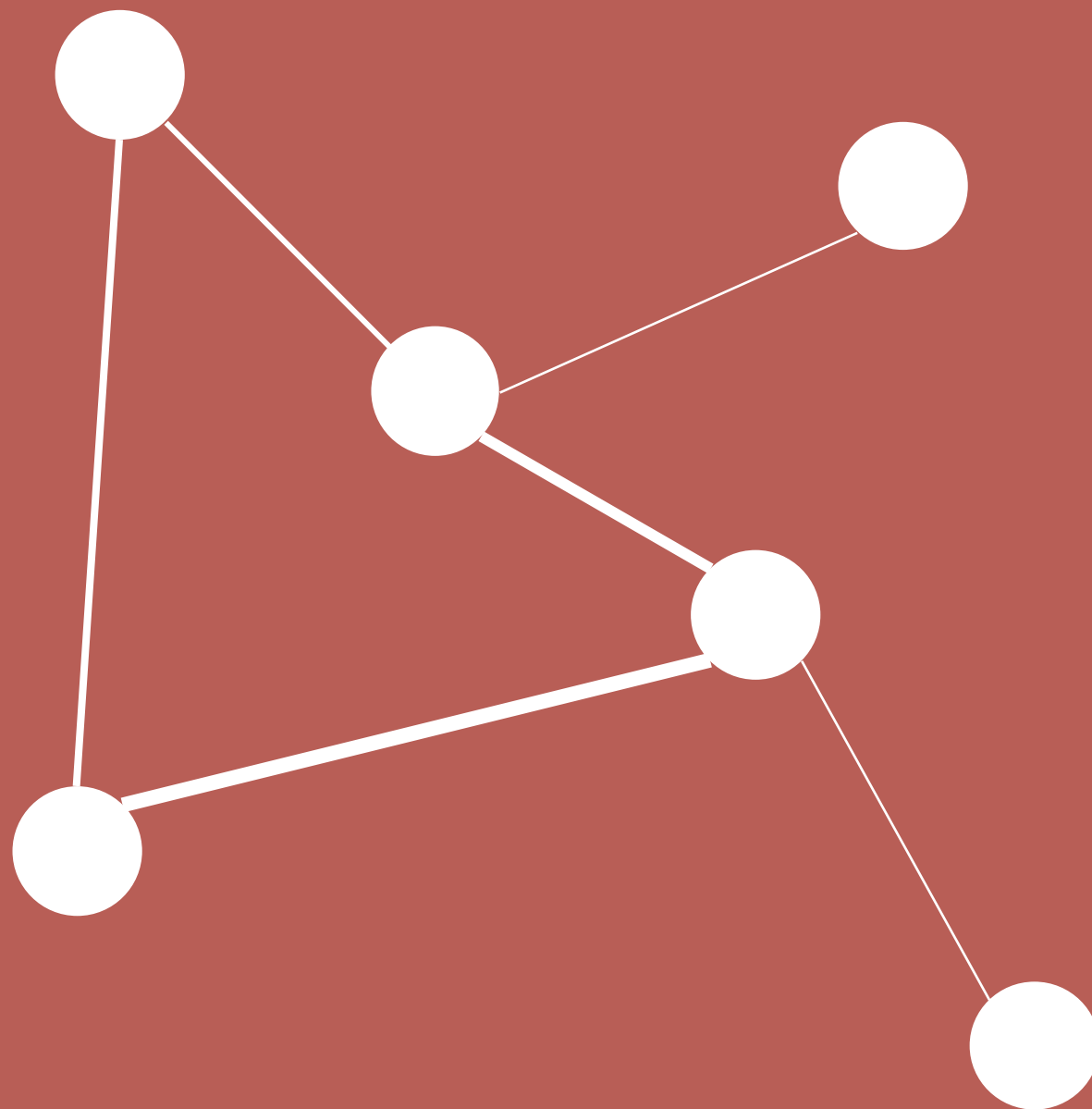


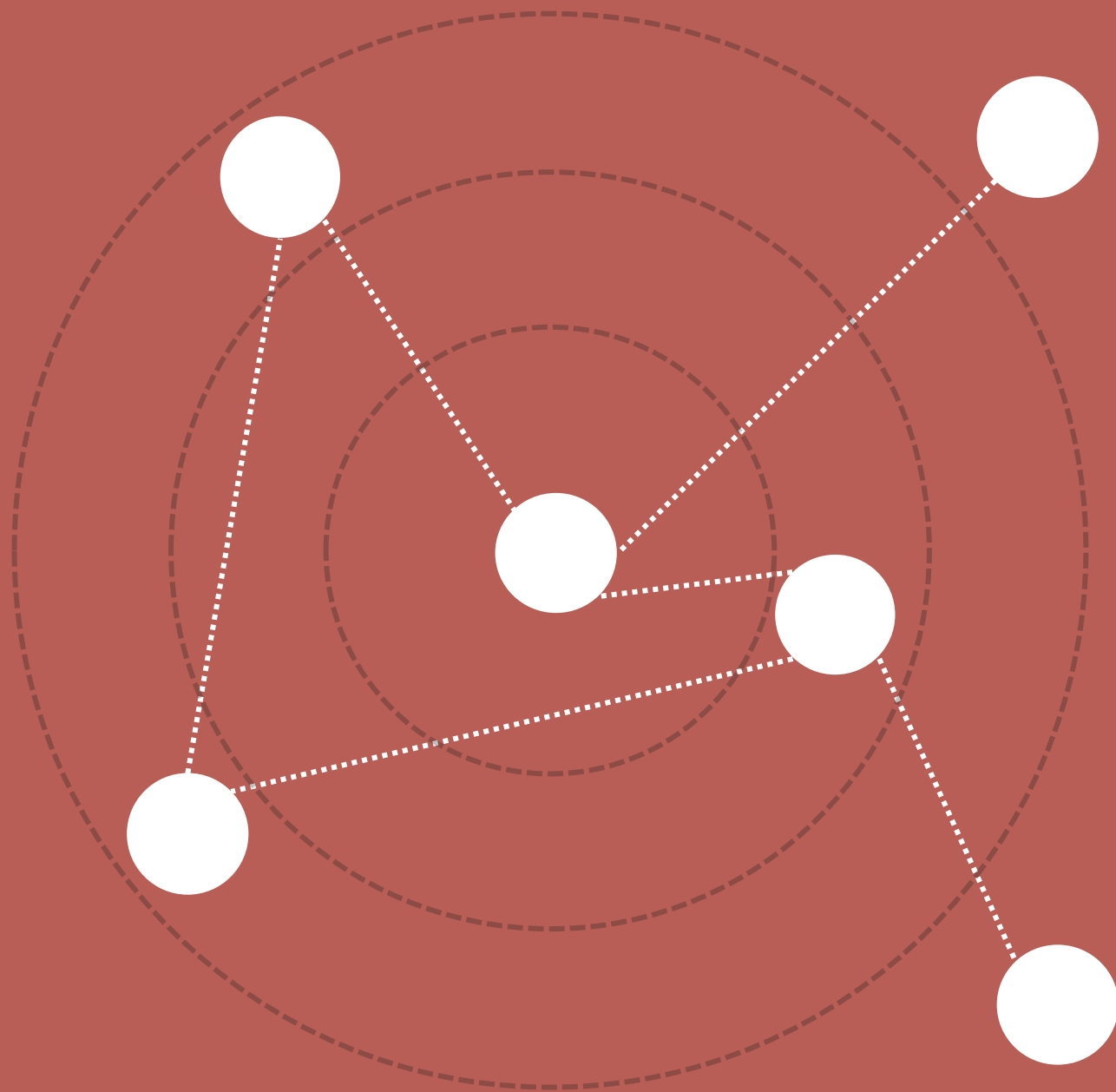












Optimization based on a set of criteria
(mathematical aesthetics):

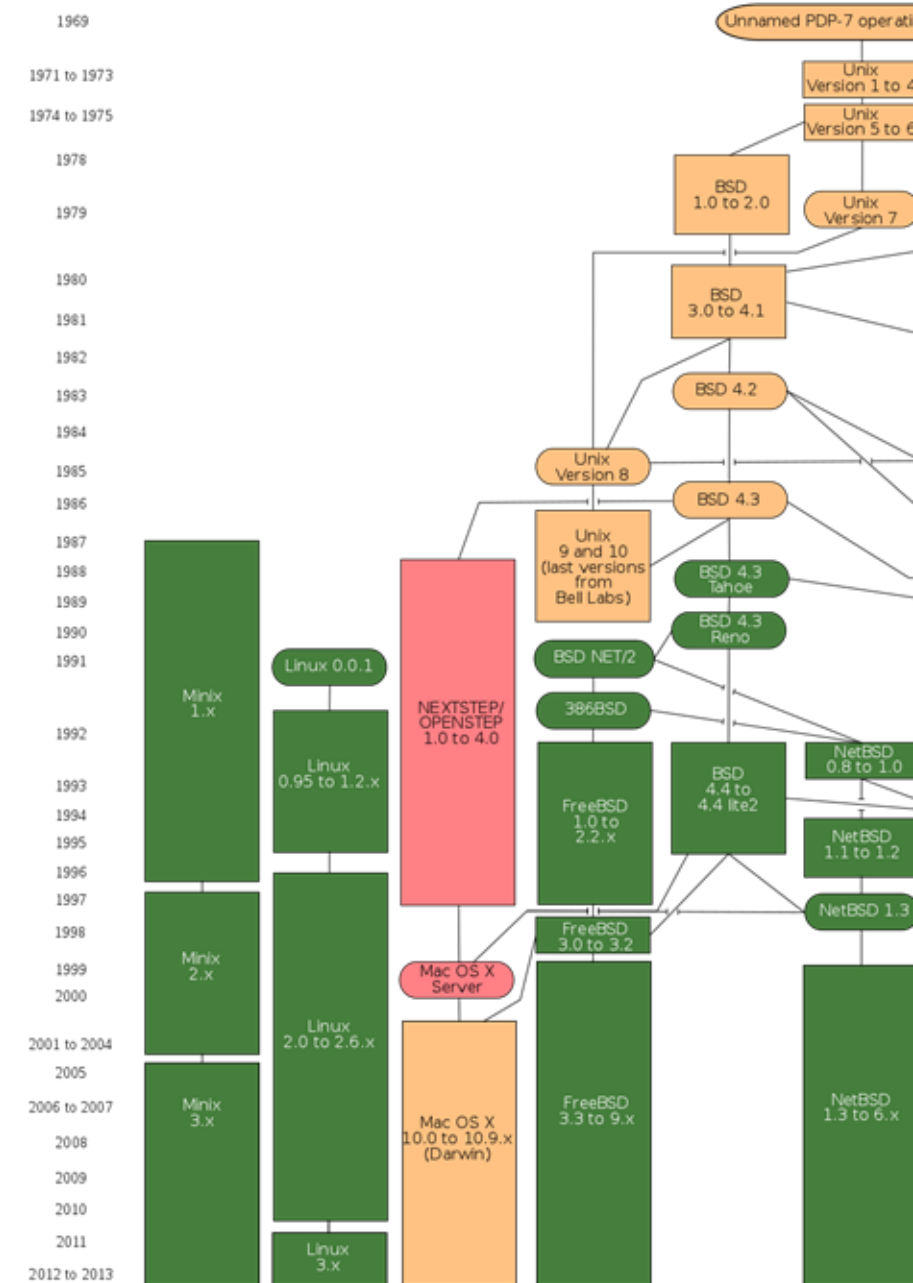
Minimize edge crossings

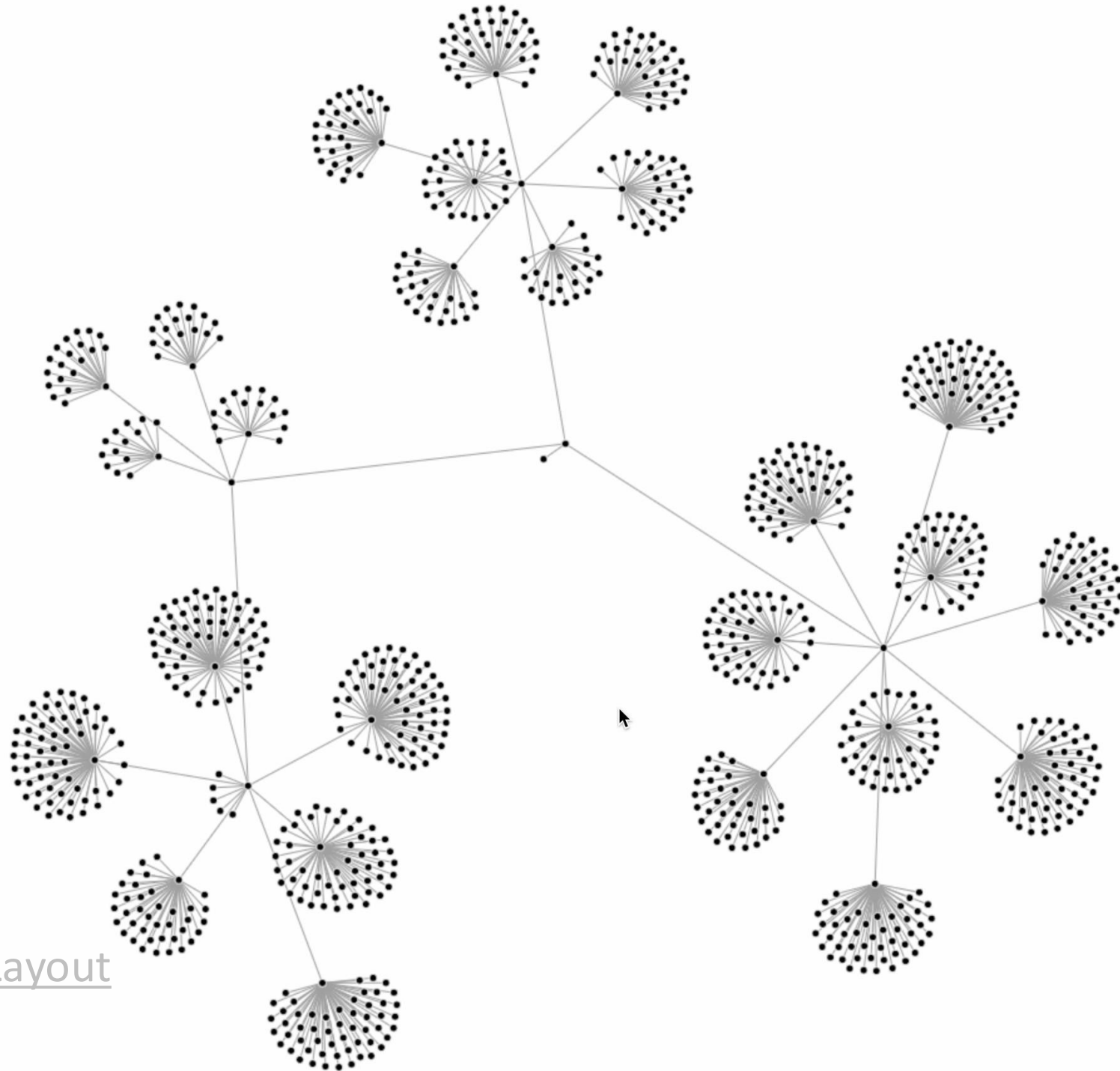
Minimize area

Maximize symmetry

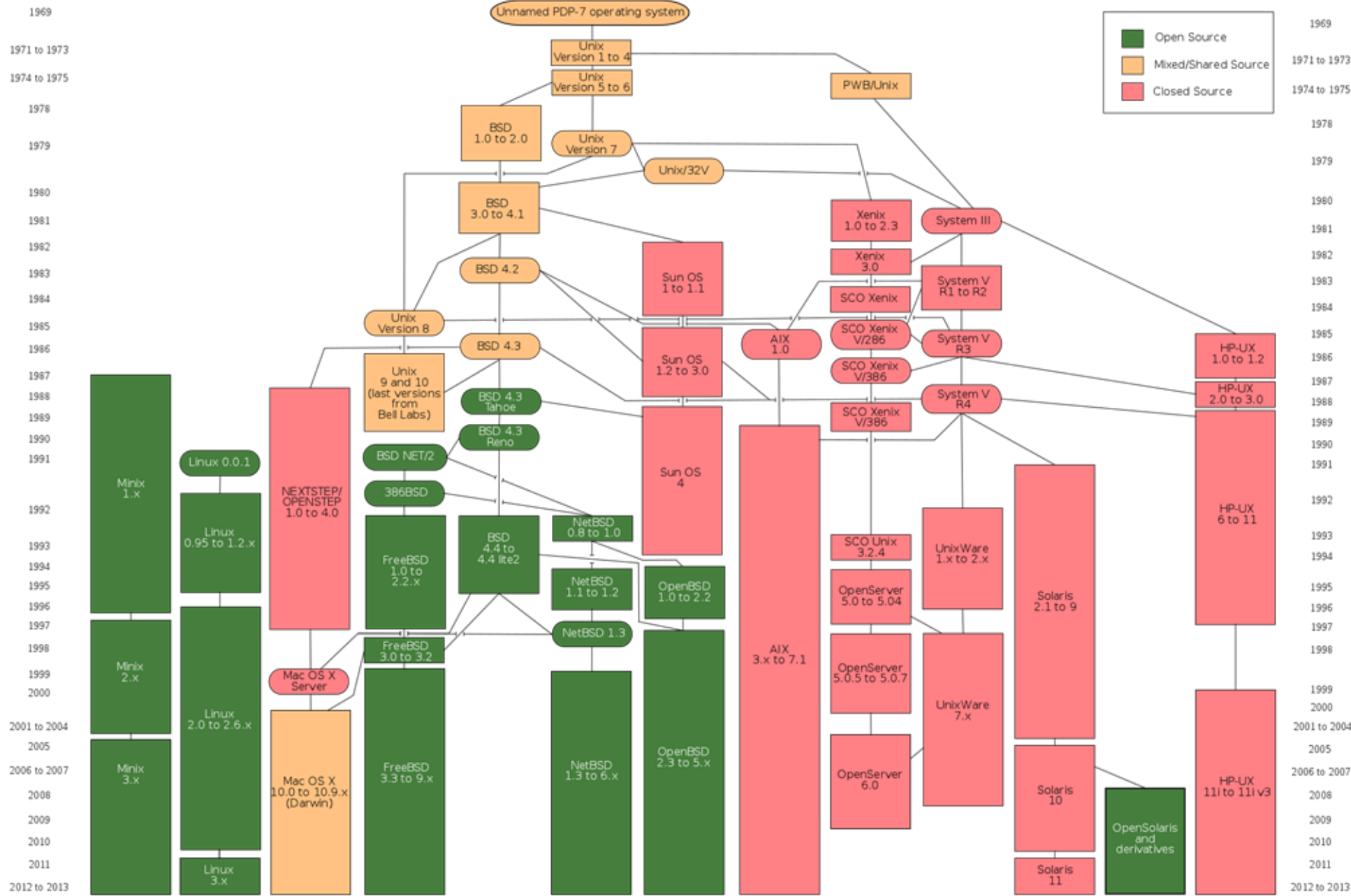
Can't do all at once!

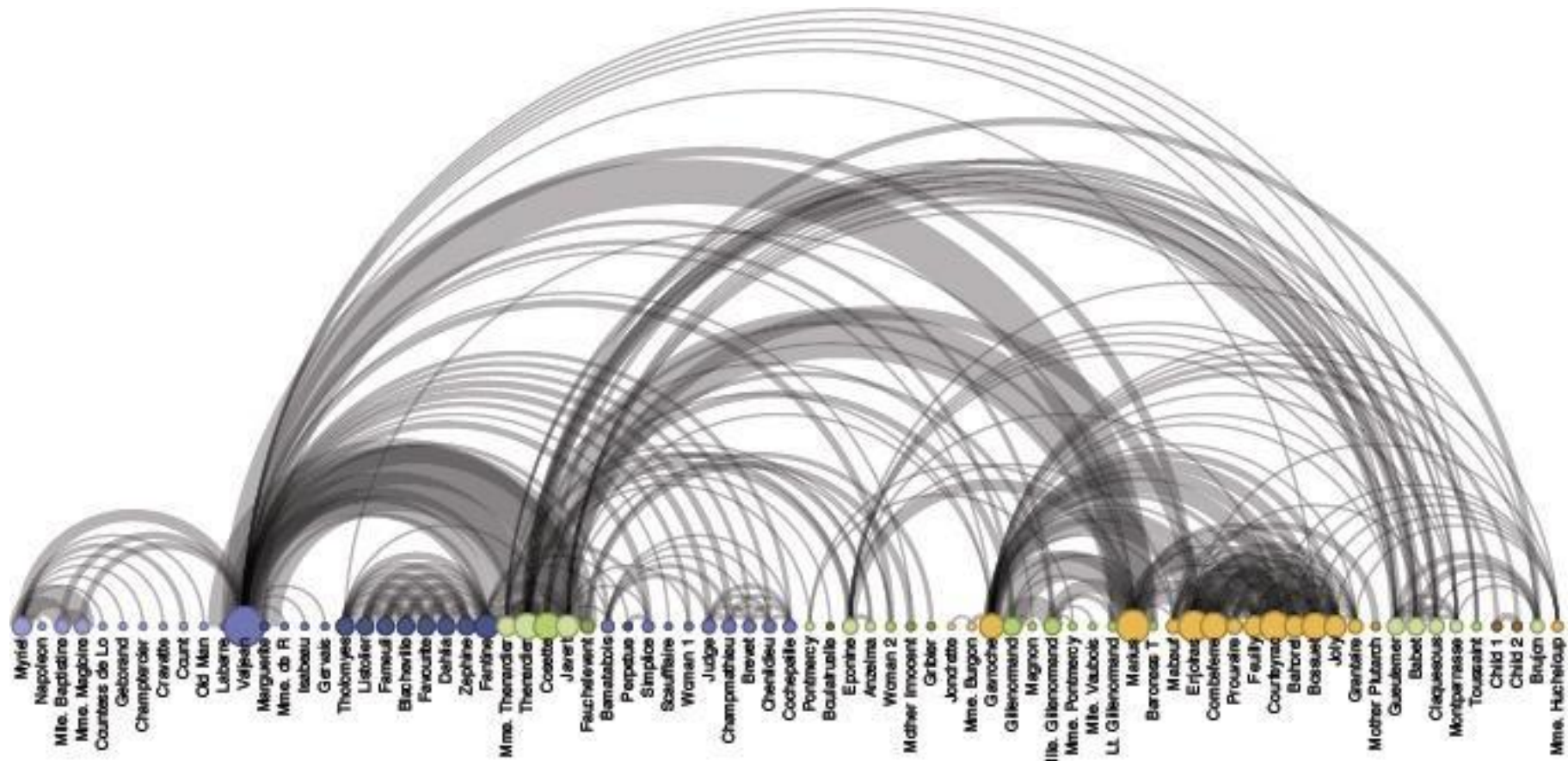
Often unsuitable for interactive
visualizations





Force Directed Layout

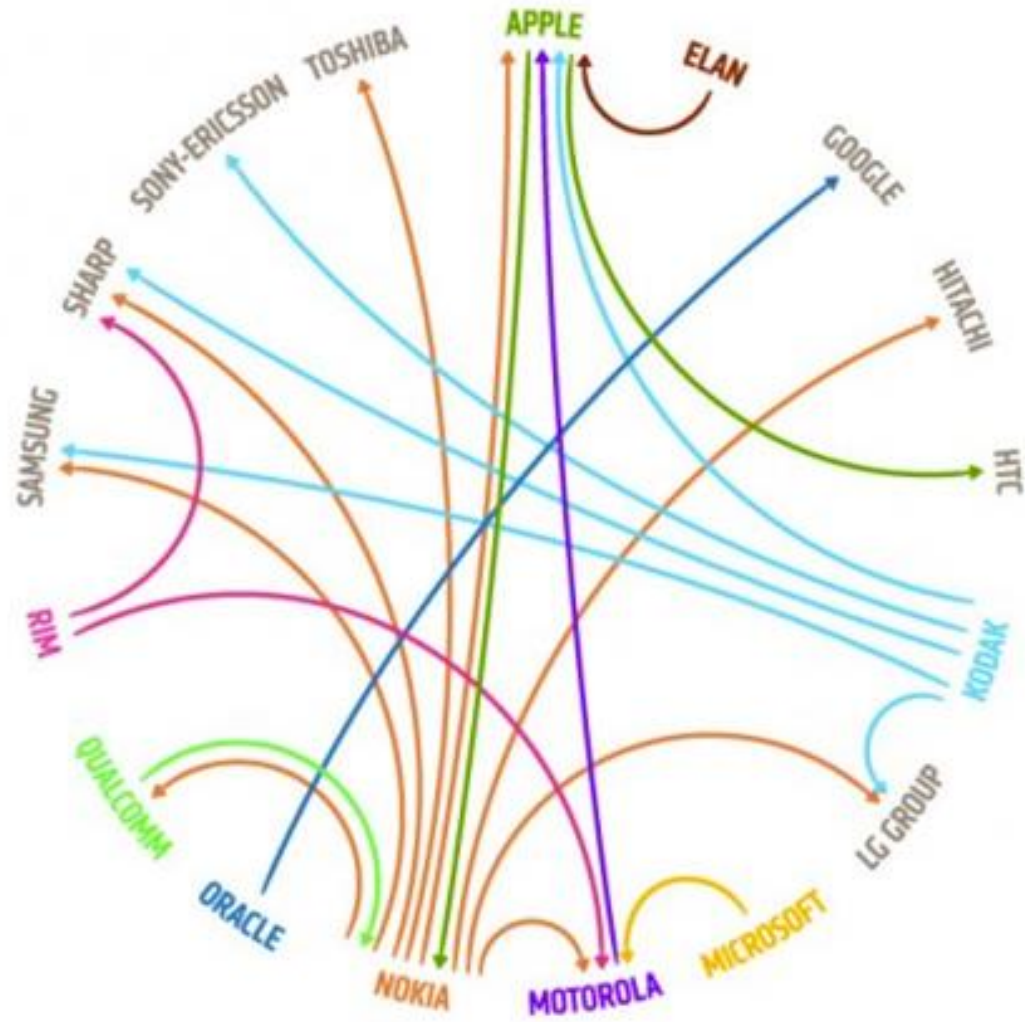




<http://hci.stanford.edu/jheer/files/zoo/ex/networks/arc.html>

Arc Diagram

LAWSUITS IN THE MOBILE BUSINESS



Radial Network

**Total
Revenue
2015**

\$5.2

trillion

Includes local, state, and
federal governments



See where the
money comes from

**Total
Spending
2015**

\$5.7

trillion

Includes local, state, and
federal governments

Establish Justice and Ensure Domestic...

Provide for the Common Defense

Promote the General Welfare

Secure the Blessings of Liberty to Ourselves
and Our Posterity

Crime and Disaster

Child Safety and Miscellaneous Social
Safeguarding Consumers and Employees

National Defense and Support for
Veterans

Foreign Affairs and Foreign Aid
Immigration and Border Security (net of...)

Standard of Living and Aid to the
Disadvantaged

Economy and Infrastructure

Health

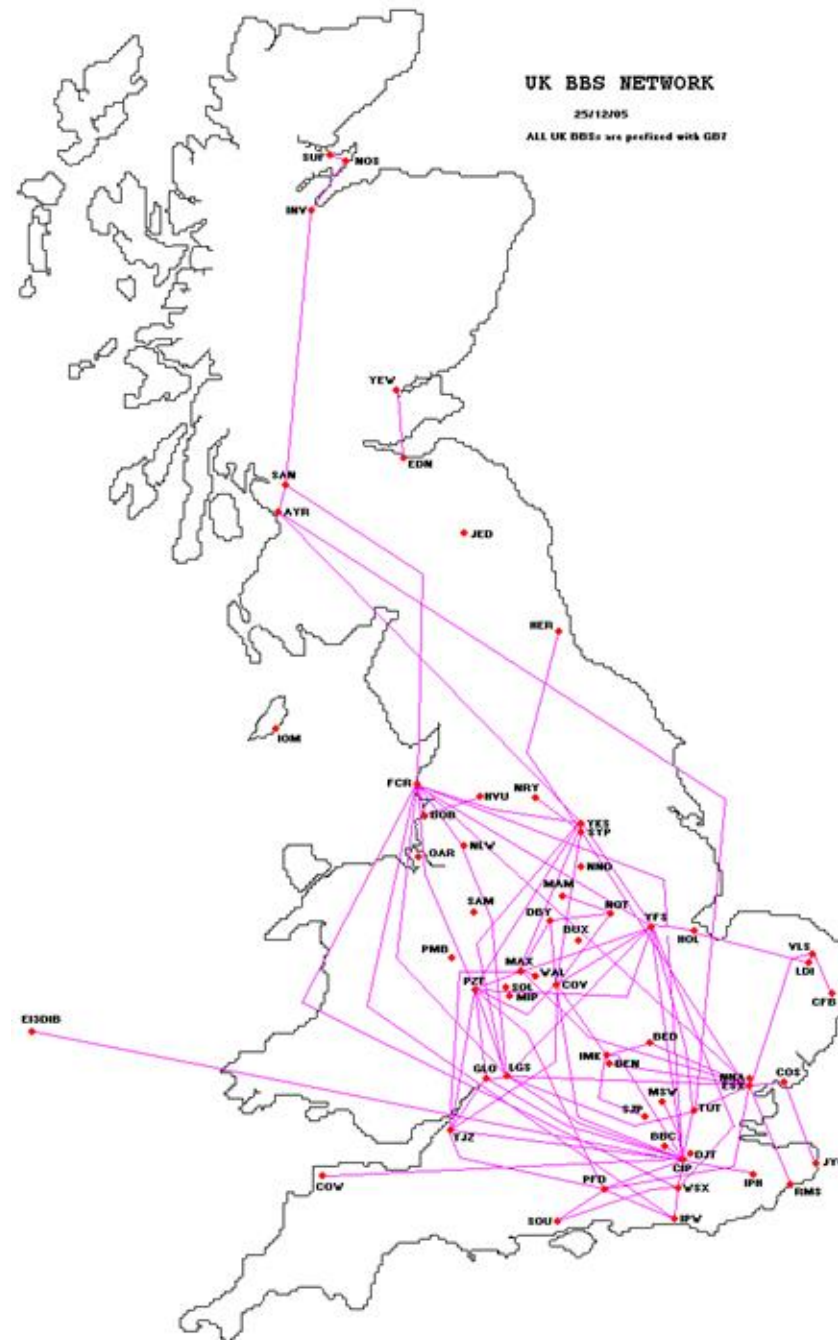
Government-Run Businesses

Wealth and Savings

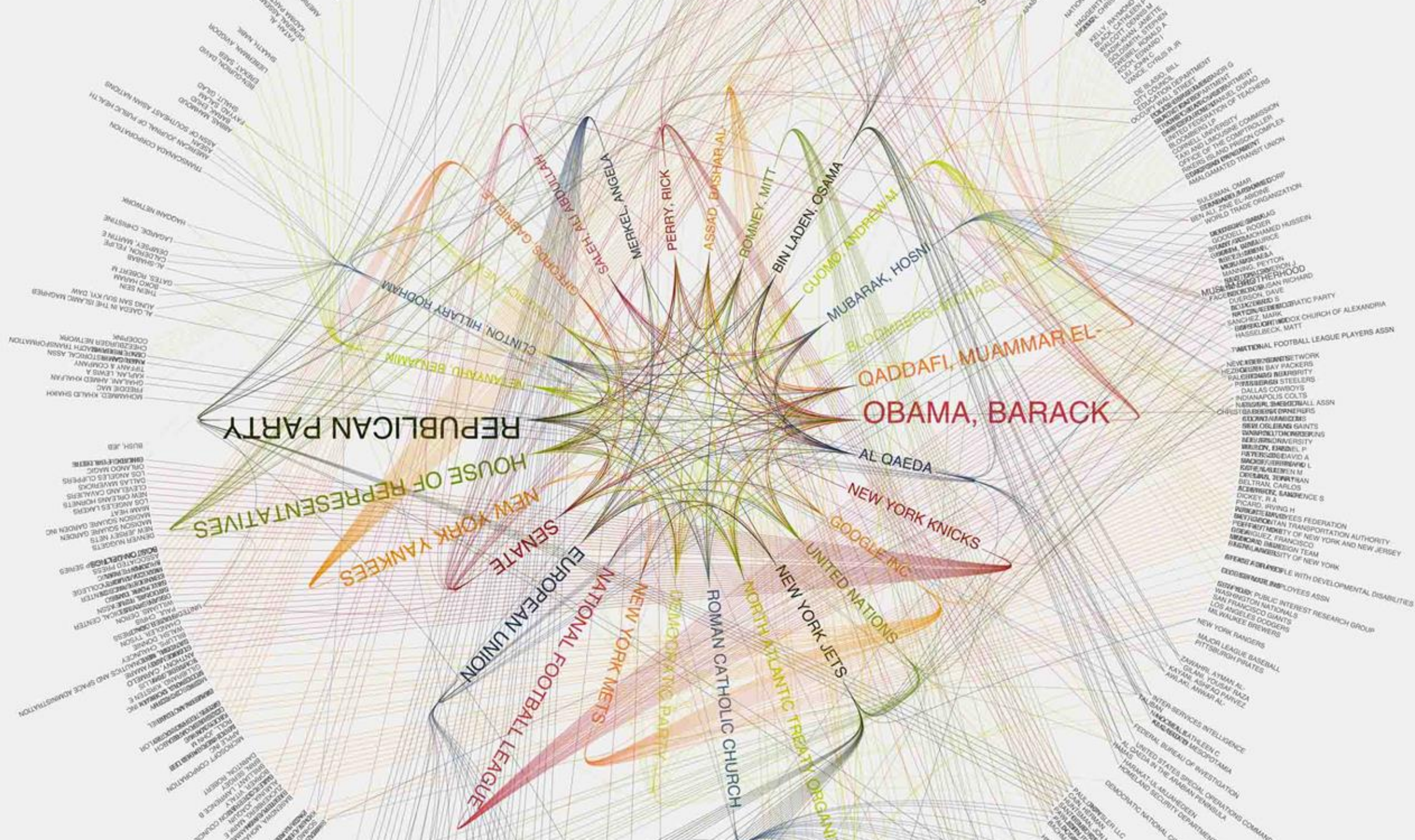


Spending
details

Sankey Diagram



Geographic Layouts



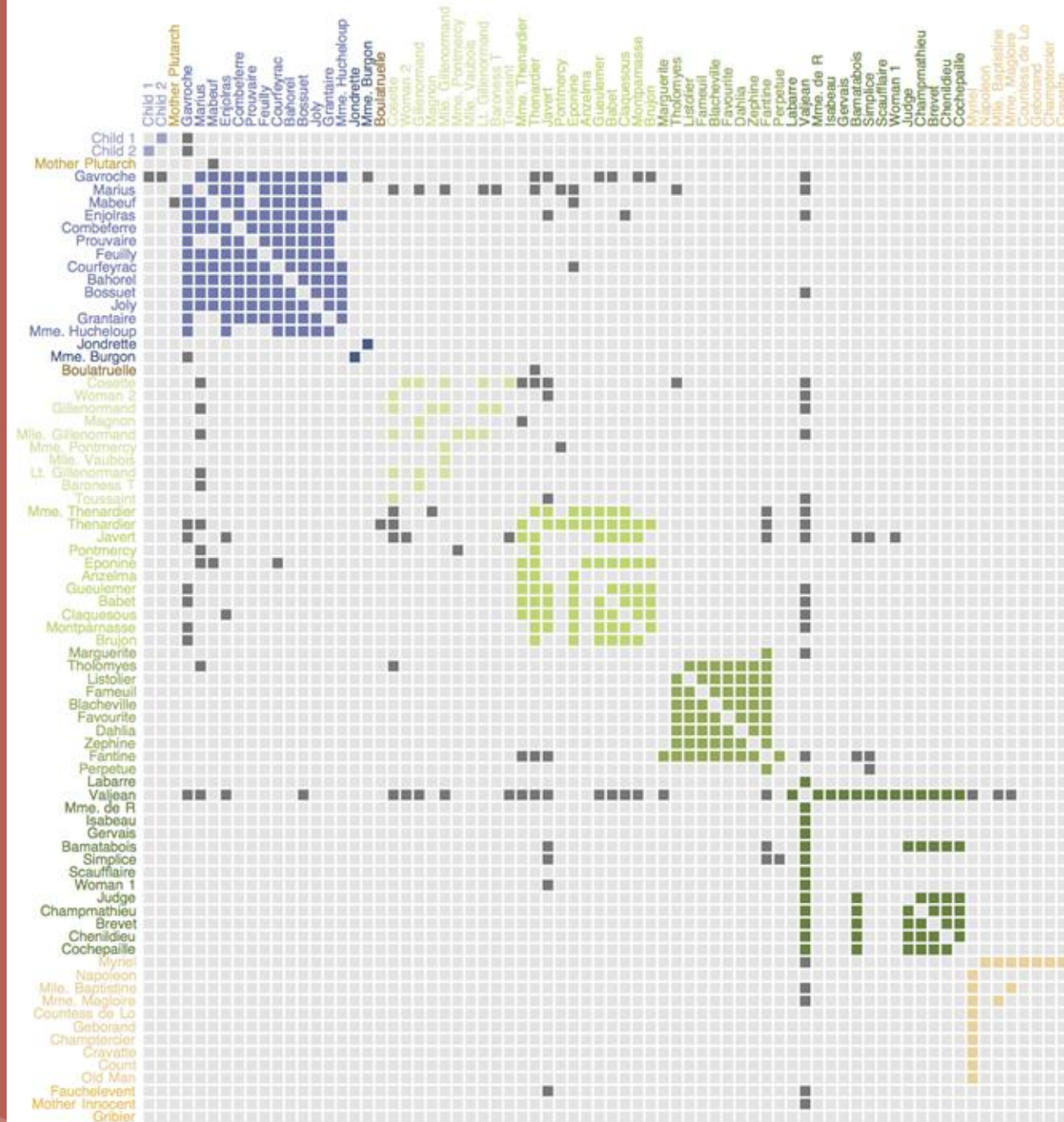
Mitigate occlusion problems

Eliminate edge-crossings

Less limited by data size

Suffer from abstraction

Adjacency Matrices



Performance considerations (layout calculation, animation)

Hard to see contents (nodes & edges may occlude)

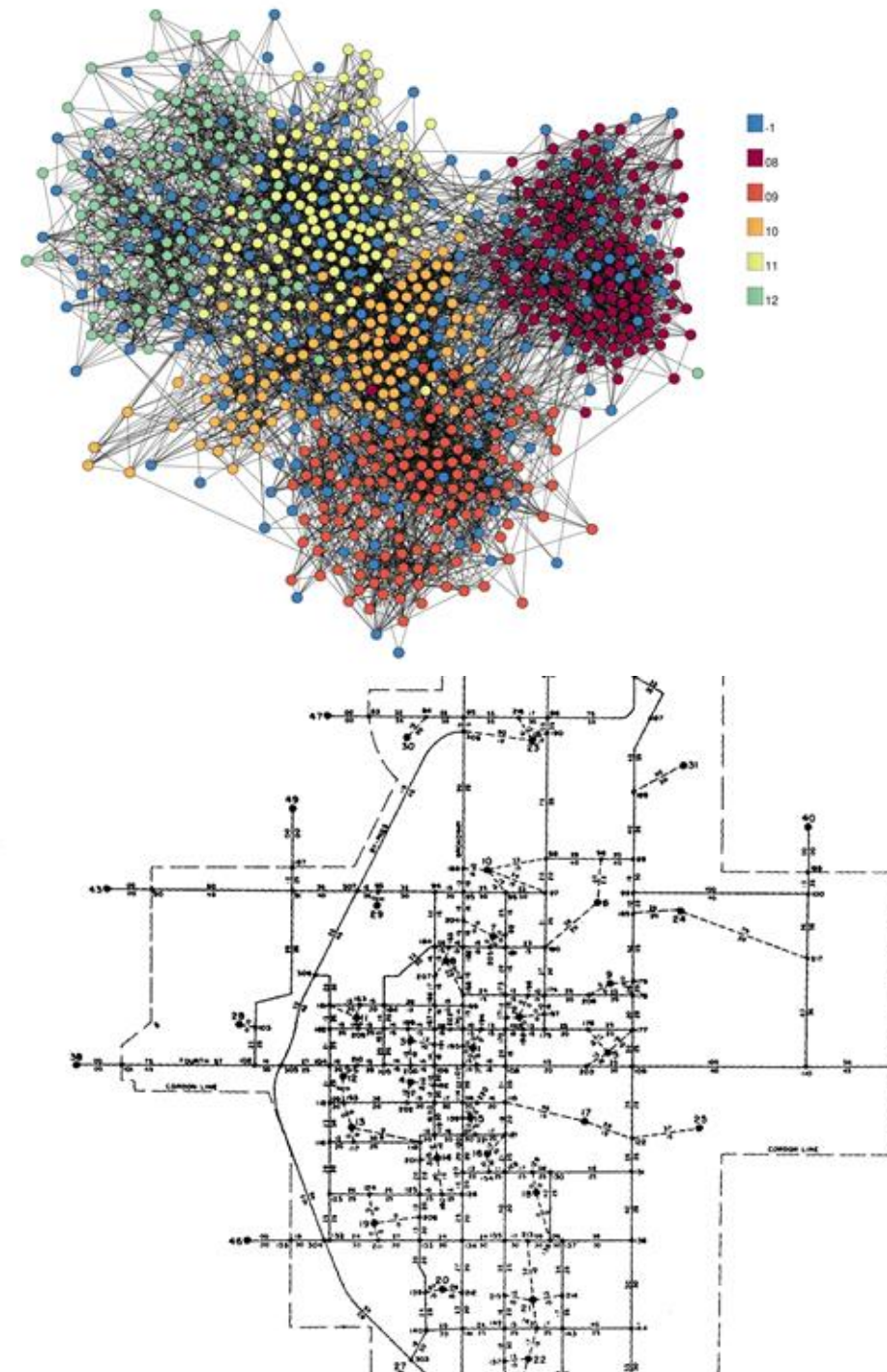
Hard to select & interact

Hard to navigate

Hard to see & understand structure

Hard to label clearly and efficiently

Challenges



Minimize edge-crossings

Distribute nodes & edges evenly

Make edges (links) have similar lengths (unless encoding data)

Display similar substructures in similar ways

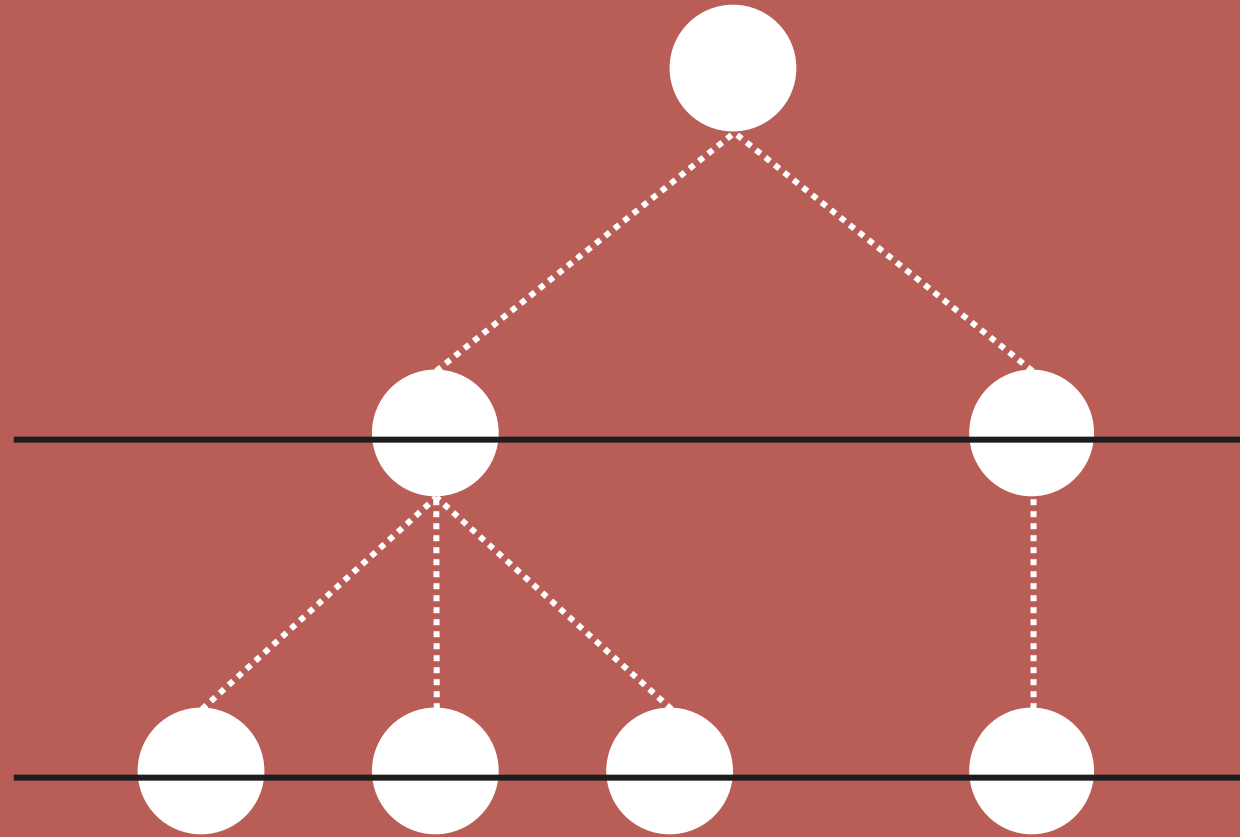
Use space efficiently

General Guidelines

Place nodes at equal depth at the same visual level.

Make distance between sibling nodes fixed.

When changing layout, animate, minimizing movement of nodes & clusters to preserve user's mental map.

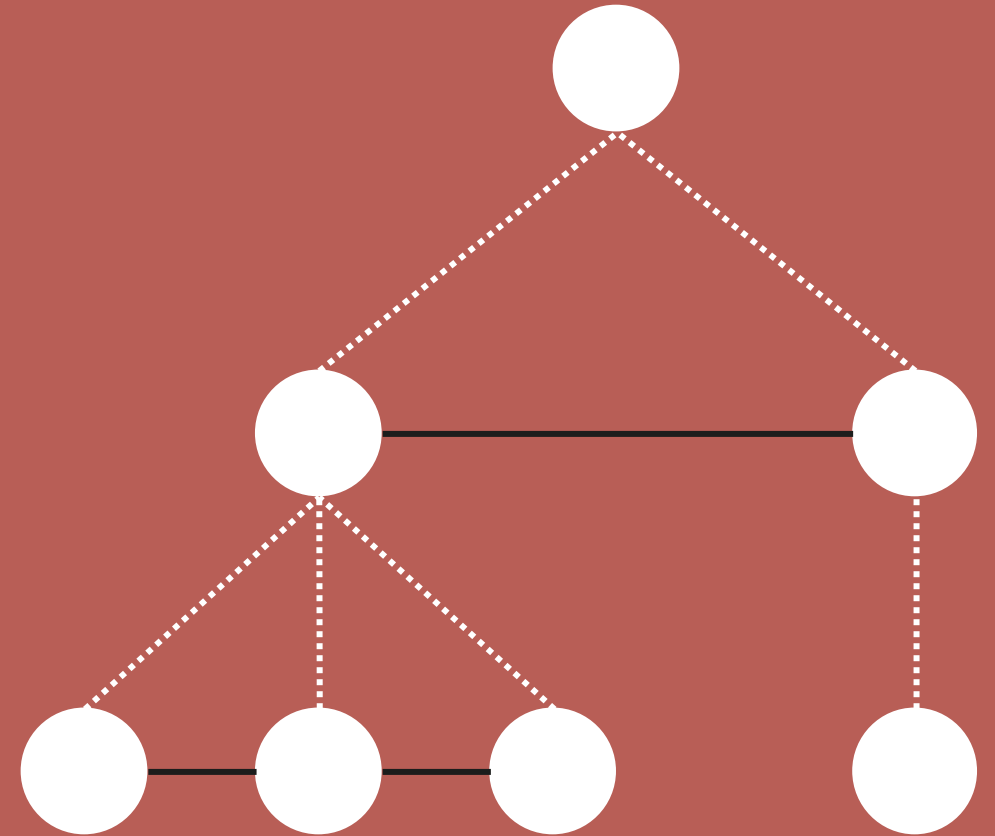


Guidelines for Trees

Place nodes at equal depth at the same visual level.

Make distance between sibling nodes fixed.

When changing layout, animate, minimizing movement of nodes & clusters to preserve user's mental map.

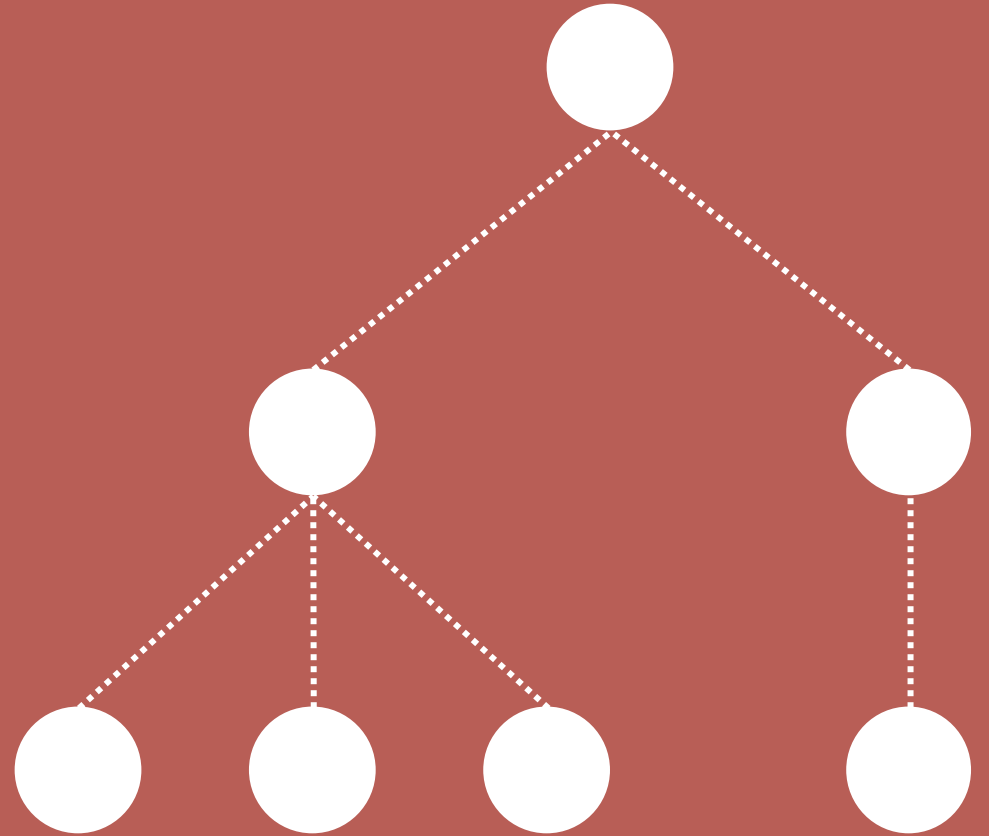


Guidelines for Trees

Place nodes at equal depth at the same visual level.

Make distance between sibling nodes fixed.

When changing layout, animate, minimizing movement of nodes & clusters to preserve user's mental map.



Guidelines for Trees

Dinner Break
until 7:45 pm

Find **one** team member from another group and ask them what they are able to understand from **one** of your visualizations **without prompting or context!**

Then after they try their best to understand it **unaided**, provide them with context and **discuss**.

Then do this **one more time (two rounds)**

15 min + 15 min
Round 1 + Round 2

Create a brief overview (5 min) of

- 1) your current project visualizations; and,**
- 2) The user questions they answer**
- 3) Any new ideas / directions**
- 4) Current challenges**

Present for 5 min, with 3 min Q&A

Project check-in: Wk 8

Group

**Use the remainder of the class time
for group work**

Group work time: Wk 8

Thanks!